CHAPTER 11

Spectroscopy

Content and rendering review packet

Original app content and actual Chromium screenshots. No textbook passages or textbook figures are embedded. Source labels and curriculum headings are not evidence of factual correctness.

Source snapshot SHA-256: ed978f297b8cf2ece50285a78a7fad18a23424167fb9b8d2941a7e662d626fe7

Review all claims, molecule graphs, lessons, exercises, hints and displayed states. Keys are author claims, not ground truth. Screenshots show static states with reduced motion: they do not prove animation timing or persistence behavior.

For every finding: item ID/page; severity (critical/major/minor); exact claim; derivation or reference evidence; proposed correction; confidence. List unreviewed/uncertain items. Model agreement is not scientific verification.

Coverage checklist

ch11:exercise:ir-read | ir-read:stage-1, ir-read:stage-2, ir-read:stage-3

ch11:exercise:uv-energy | uv-energy:stage-1, uv-energy:stage-2, uv-energy:stage-3

ch11:exercise:nmr-ethyl | nmr-ethyl:stage-1, nmr-ethyl:stage-2, nmr-ethyl:stage-3

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ch11:lesson:ir | ir:step-1, ir:step-2, ir:step-3

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ch11:molecule:acetone | acetone

ch11:molecule:isopropanol | isopropanol

ch11:molecule:aceticacid | aceticacid

ch11:molecule:butadiene | butadiene

ch11:molecule:ethene | ethene

ch11:molecule:ethylacetate | ethylacetate

ch11:molecule:propanal | propanal

ch11:curriculum | scope-1, scope-2, scope-3

Curriculum scope

scope-1: 11.1 Infrared Spectroscopy

scope-2: 11.2 Ultraviolet Spectroscopy

scope-3: 11.3 Nuclear Magnetic Resonance Spectroscopy

Objective: Recognize approximate IR regions

Objective: Relate conjugation, photon energy, and absorbance

Objective: Interpret NMR shifts, integrals, and splitting

Supplied Kaplan curriculum: Chapter 11; section scope checked against local body and summary | organic_chemistry_kaplan/OEBPS/chapter11b.xhtml

IR functional-group regions: OpenStax 12.8 | https://openstax.org/books/organic-chemistry/pages/12-8-infrared-spectra-of-some-common-functional-groups

First-order proton splitting: OpenStax 13.6 | https://openstax.org/books/organic-chemistry/pages/13-6-spin-spin-splitting-in-1h-nmr-spectra

Conjugation and UV absorption: OpenStax 14.8 | https://openstax.org/books/organic-chemistry/pages/14-8-interpreting-ultraviolet-spectra-the-effect-of-conjugation

NMR-active nuclei: OpenStax 13.1, inspected | https://openstax.org/books/organic-chemistry/pages/13-1-nuclear-magnetic-resonance-spectroscopy

UV HOMO/LUMO excitation: OpenStax 14.7, inspected | https://openstax.org/books/organic-chemistry/pages/14-7-ultraviolet-spectroscopy

No source passage is included. Request relevant authoritative excerpts when a claim cannot be checked by derivation.

ir | IR: recognize functional groups

Vibrations must change dipole moment

ir:rule | Core Rules

IR probes molecular vibrations. A vibration absorbs IR when it changes the molecular dipole moment. Useful approximate regions: alcohol O–H 3200–3600 cm^{-1} (often broad); acid O–H 2500–3300 (very broad); C=O about 1650–1800; nitrile about 2210–2260. N–H is often near 3300, generally less broad than hydrogen-bonded OH.

Visual card: Infrared (IR) Spectroscopy: Hooke's Law (Bond Strength vs. Reduced Mass) & Key Diagnostic Zones

Chemistry diagram: Matched molecule and illustrative IR bands. Band shapes and heights are not experimental data.

$\tilde{\nu}$ (cm^{-1}) $\propto \sqrt{(k / \mu)}$ → Stronger Bond (↑ k: Triple > Double > Single) & Lighter Atoms (↓ μ: O–H > C–C) = HIGHER Wavenumber!

O–H: 3200–3600 cm^{-1} (Broad)

N–H: 3300–3500 cm^{-1} (Sharp; 1°=2 Spikes)

C=O: 1700–1750 cm^{-1} (Strong Sharp)

sp C–H ~3300 | C≡N/C≡C ~2200

Alcohol O–H | 3200–3600 cm^{-1} (Smooth U-shaped broad) | Carboxylic acid O–H: 2500–3300 cm^{-1} (ultra-broad beard)

Amine / Amide N–H | 3300–3500 cm^{-1} (Medium, sharper than OH) | 1° NH_2 = 2 peaks; 2° NH = 1 peak; 3° N = 0 peaks

Carbonyl C=O | 1650–1810 cm^{-1} (Very strong, sharp) | Conjugation ↓ wavenumber by ~20–30 cm^{-1}; Ring strain ↑ wavenumber

Requirement for IR Absorption — — A vibration is IR-active ONLY if it changes the molecule's net dipole moment (symmetric stretches of O_2, N_2, or trans-2-butene C=C are IR-inactive!).

Four Clean IR Zones — — Zone 1 (2500–3600): O–H, N–H, C–H | Zone 2 (2100–2260): C≡C, C≡N | Zone 3 (1600–1820): C=O, C=N, C=C | Fingerprint (<1500): C–O (1050–1250).

ir:cue | Memory Cues

Use a band pattern, not one magic number. The fingerprint region below about 1500 cm^{-1} helps comparison.

Visual card: 3-Second IR Functional Group Identifier

What signature peak do you see in the IR spectrum?

Strong Sharp Peak Around 1700–1750 cm^{-1} — Carbonyl (C=O) Present! Now Check 2500–3600 cm^{-1}: — + Broad 2500–3300 = Carboxylic Acid; + 2720/2820 doublet = Aldehyde; + 1650/3350 = Amide; + 1735 & C–O 1200 = Ester; Alone at 1715 = Ketone!

Broad Rounded Trough at 3200–3600 cm^{-1} (No 1700 Peak) — Alcohol (O–H) or Phenol! — Intermolecular hydrogen bonding creates a wide distribution of O–H bond strengths, broadening the peak into a smooth U-shape.

Conjugation vs. Ring Strain on C=O Wavenumber — Conjugation LOWERS $\tilde{\nu}$ (~1685 cm^{-1}); Small Ring Strain RAISES $\tilde{\nu}$ (~1780 cm^{-1}) — Conjugation gives C=O more single-bond character (↓ k → ↓ cm^{-1}); small 4/5-membered rings force higher s-character into the exocyclic C=O (↑ k → ↑ cm^{-1}).

ir:trap | Common Mistakes / MCAT Traps

Carbonyl positions vary with functional group, conjugation, ring strain, and environment. Absence of an OH band may support a structure, but sample conditions matter. IR alone seldom gives complete connectivity.

Visual card: Confusing Alcohol O–H (3350 cm^{-1}) with Carboxylic Acid O–H (2500–3300 cm^{-1} + 1710 cm^{-1})

Common Trap — — Seeing an O–H peak without a 1700 cm^{-1} C=O peak and calling it a carboxylic acid, or thinking symmetric alkyne C≡C stretches are strong.

Correct Reasoning — — A carboxylic acid MUST have BOTH a strong C=O peak (~1710 cm^{-1}) AND a massive broad O–H beard (2500–3300 cm^{-1}); without ~1700 cm^{-1}, an O–H peak is an alcohol/phenol!

ir:app | MCAT Applications

Follow loss of a carbonyl band and appearance of OH when monitoring a carbonyl reduction, together with other evidence.

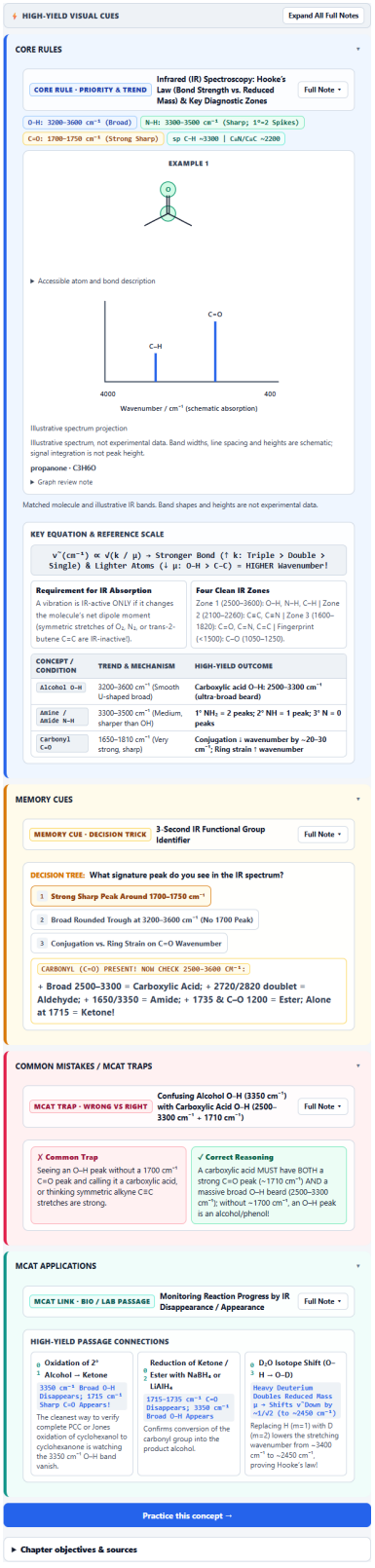
Visual card: Monitoring Reaction Progress by IR Disappearance / Appearance

Oxidation of 2° Alcohol → Ketone — 3350 cm^{-1} Broad O–H Disappears; 1715 cm^{-1} Sharp C=O Appears! — The cleanest way to verify complete PCC or Jones oxidation of cyclohexanol to cyclohexanone is watching the 3350 cm^{-1} O–H band vanish.

Reduction of Ketone / Ester with $NaBH_4$ or $LiAlH_4$ — 1715–1735 cm^{-1} C=O Disappears; 3350 cm^{-1} Broad O–H Appears — Confirms conversion of the carbonyl group into the product alcohol.

D_2O Isotope Shift (O–H → O–D) — Heavy Deuterium Doubles Reduced Mass μ → Shifts $\tilde{\nu}$ Down by ~1/√2 (to ~2450 cm^{-1}) — Replacing H (m=1) with D (m=2) lowers the stretching wavenumber from ~3400 cm^{-1} to ~2450 cm^{-1}, proving Hooke's law!

Screenshot: ir-visuals



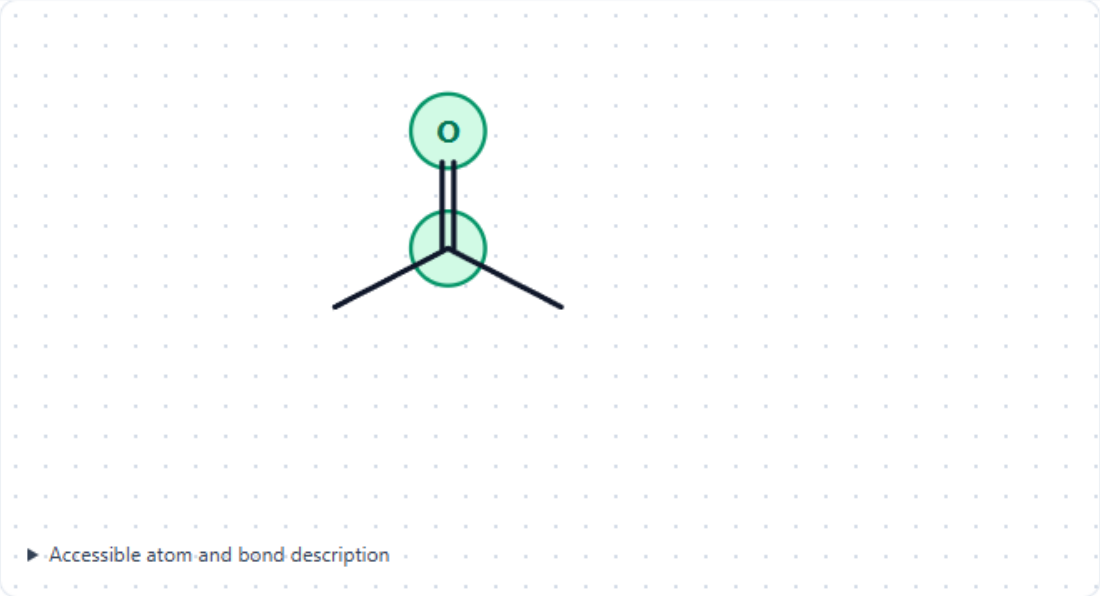
ir:step-1 | 1. Carbonyl signature

A simple saturated acyclic ketone often absorbs near 1715 cm^{-1}. The illustrative plot marks a band position, not an experimentally measured spectrum.

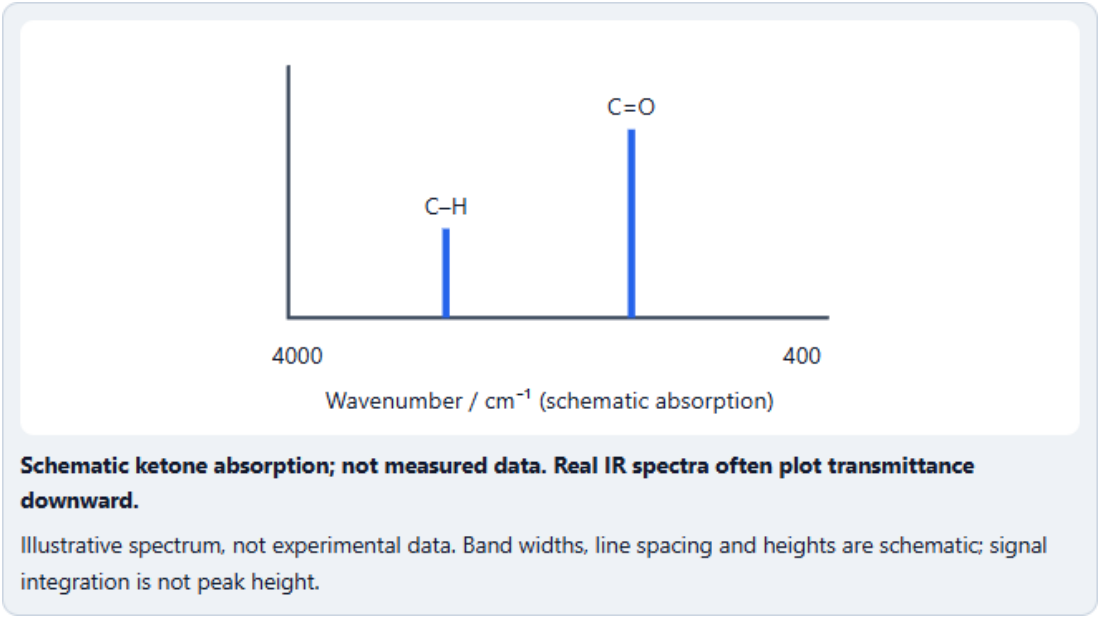
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Screenshot: ir-step-1



Screenshot: ir-step-1-diagram



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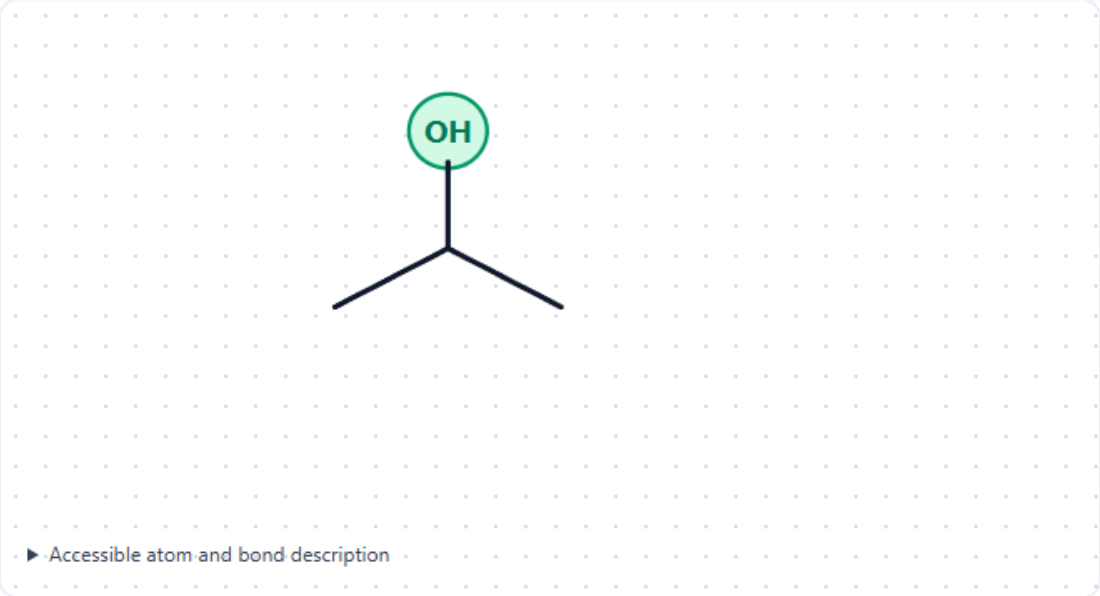
ir:step-2 | 2. Compare OH

An alcohol has an O–H vibration often broadened by hydrogen bonding. Converting acetone to propan-2-ol changes both the C=O and O–H evidence.

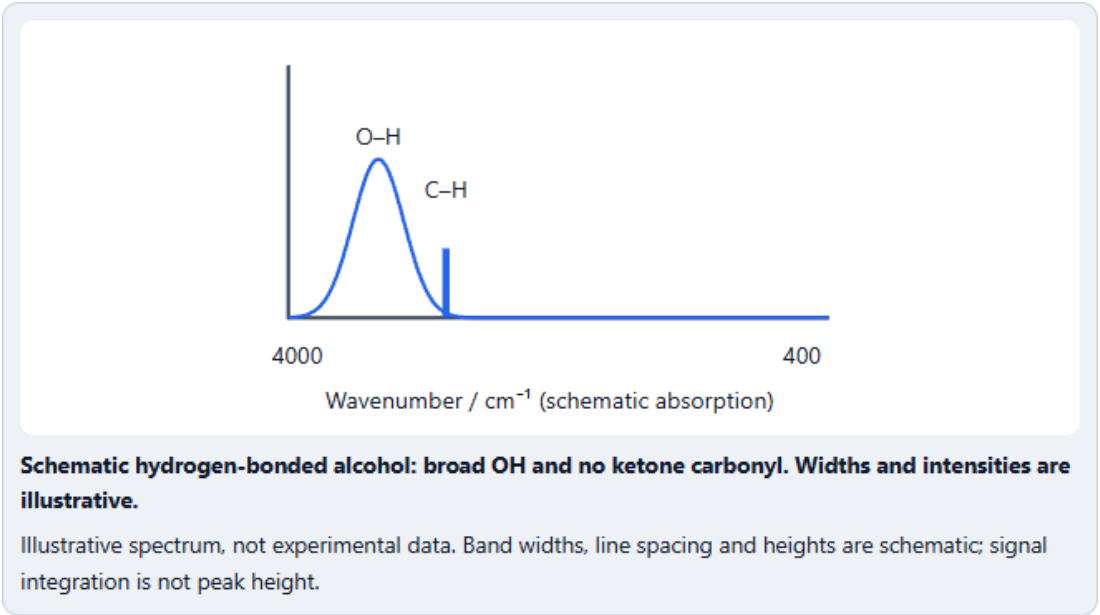
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Screenshot: ir-step-2



Screenshot: ir-step-2-diagram



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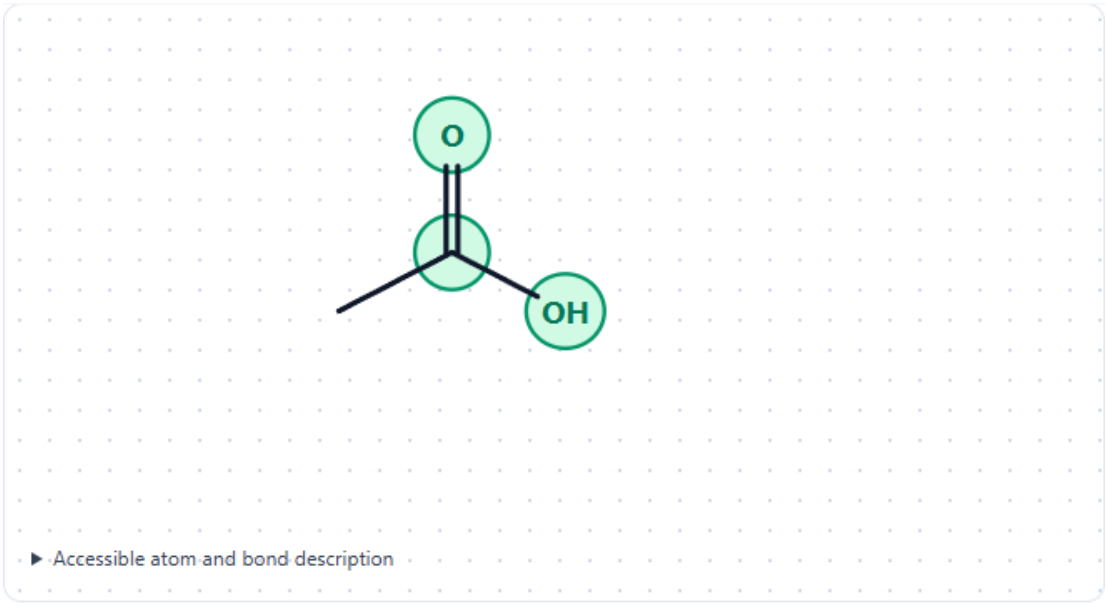
ir:step-3 | 3. Acid pattern

A carboxylic acid can show both a carbonyl band and a very broad OH band extending across roughly 2500–3300 cm^{-1}. Interpret them together.

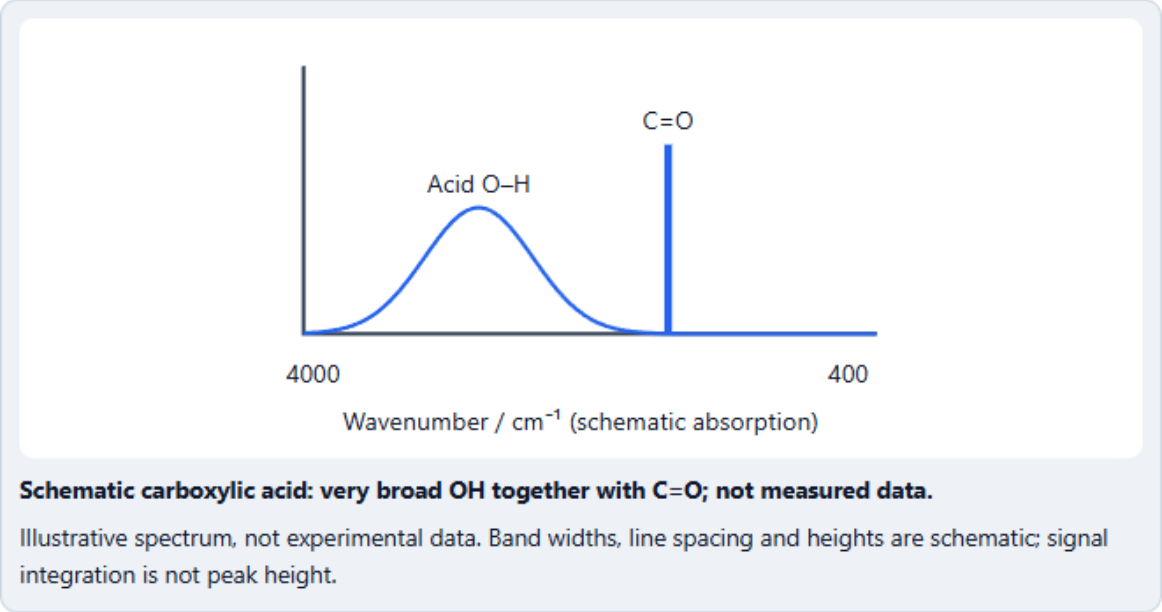
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Screenshot: ir-step-3



Screenshot: ir-step-3-diagram



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uv | UV–visible: conjugation and energy

Longer wavelength means lower photon energy

uv:rule | Core Rules

UV–visible absorption promotes electrons between electronic states. In many comparable conjugated pi systems, increasing conjugation lowers the transition energy and shifts absorption to longer wavelength. Photon energy obeys E=hc/λ. Beer–Lambert law is A=εlc in its linear regime. For a simple pi system, a pi → pi* excitation can promote an electron from the highest occupied molecular orbital (HOMO) to the lowest unoccupied molecular orbital (LUMO). Extended conjugation often narrows that gap in comparable systems; allowed transitions and actual spectra also depend on the molecule.

Visual card: UV–Visible Spectroscopy: HOMO → LUMO Gap, Conjugated Polyenes & Complementary Colors

Chemistry diagram: Conjugated and isolated pi-system examples. No numerical wavelength or universal rate effect is inferred.

ΔE = hν = hc/λ; A = εcl. Conjugation often lowers the relevant transition energy; there is no universal wavelength increment per double bond.

π → π* and n → π* Transitions

↑ Conjugation → ↓ ΔE → ↑ λ_max (Bathochromic Shift)

A = ε × c × l (Beer-Lambert)

Isolated Alkene (Ethene) | 1 π bond (Large HOMO–LUMO ΔE) | λ_max = 171 nm (Vacuum UV, invisible)

1,3-Butadiene (2 Conjugated π) | Smaller HOMO–LUMO energy gap | λ_max = 217 nm (+46 nm red shift)

β-Carotene (11 Conjugated π) | Tiny HOMO–LUMO gap → absorbs blue-green (455 nm) | Reflects complementary ORANGE-RED color!

Why Conjugation Increases λ_max (Red / Bathochromic Shift) — — Combining more p orbitals into a conjugated π system raises the HOMO energy and lowers the LUMO energy, shrinking ΔE so longer-wavelength (lower-energy) photons are absorbed.

Absorbed vs. Perceived Color — — A compound that absorbs in the visible region (400–700 nm) appears as the COMPLEMENTARY color of the wavelength it absorbs (absorbs blue ~450 nm → looks orange/yellow).

uv:cue | Memory Cues

More extended conjugation often gives a red shift. Doubling wavelength halves photon energy.

Visual card: UV–Vis Conjugation & Color Predictor

What UV-Vis question is on the exam?

Rank Molecules by Increasing λ_max — Count Alternating Single-Double (Conjugated) Bonds! — More continuous conjugated π bonds = smaller ΔE = LONGER λ_max (ignore isolated double bonds separated by ≥2 sp^3 single bonds!).

What Happens to Color If a Polyene Is Bleached (O_3 or Br_2)? — Breaks Conjugation → ↑ ΔE → ↓ λ_max into UV (<300 nm) → Becomes Colorless! — Oxidative cleavage or halogenation interrupts the p-orbital highway, shifting absorption out of the visible spectrum.

pH-Sensitive Indicators (Phenolphthalein / Azo Dyes) — Deprotonation Extends Auxochrome Lone-Pair Delocalization → Shifts λ_max! — Deprotonating a phenolic –OH to $–O^-$ pushes electron density across the conjugated rings, shrinking ΔE and turning the solution bright pink.

uv:trap | Common Mistakes / MCAT Traps

Conjugation trends compare appropriate related systems, not every possible molecule. UV absorption does not uniquely identify a structure. Beer–Lambert requires compatible units and an appropriate concentration range.

Visual card: Counting Isolated Double Bonds as Conjugated or Thinking Larger ΔE Gives Larger λ_max

Common Trap — — Saying 1,4-pentadiene absorbs at a longer wavelength than 1,3-butadiene, or saying ↑ ΔE means ↑ λ_max.

Correct Reasoning — — 1,4-pentadiene has an sp^3 CH_2 group blocking p-orbital overlap (behaves like an isolated alkene, λ_max ~178 nm); energy ΔE and wavelength λ are INVERSELY proportional (↓ ΔE = ↑ λ_max)!

uv:app | MCAT Applications

Conjugated chromophores underlie retinal photochemistry and many biological absorbance measurements.

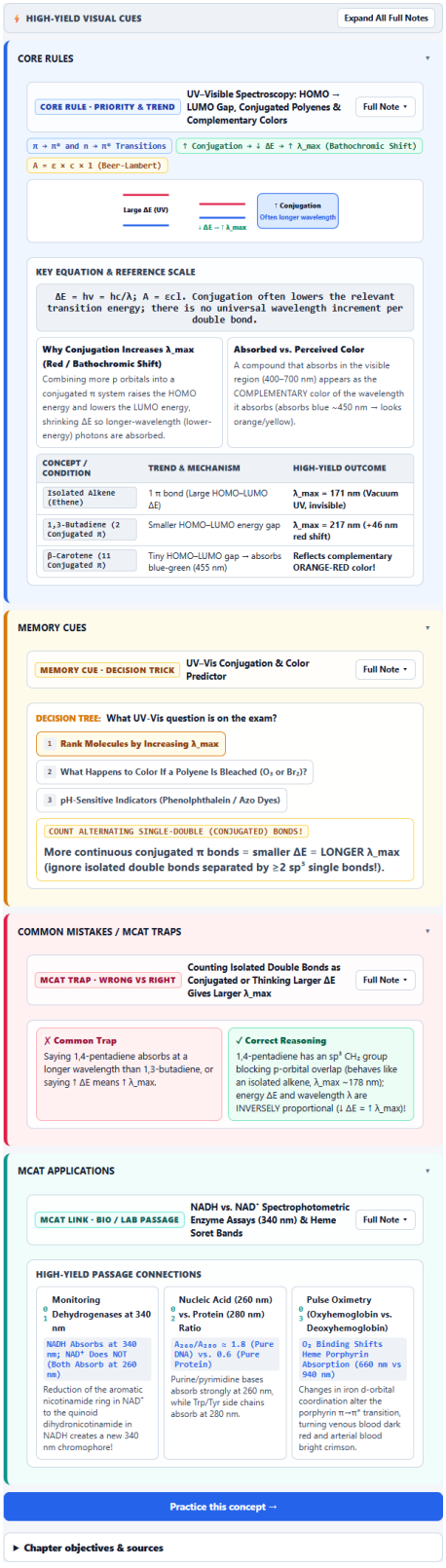
Visual card: NADH vs. NAD^+ Spectrophotometric Enzyme Assays (340 nm) & Heme Soret Bands

Monitoring Dehydrogenases at 340 nm — NADH Absorbs at 340 nm; NAD^+ Does NOT (Both Absorb at 260 nm) — Reduction of the aromatic nicotinamide ring in NAD^+ to the quinoid dihydronicotinamide in NADH creates a new 340 nm chromophore!

Nucleic Acid (260 nm) vs. Protein (280 nm) Ratio — A_{260}/A_{280} ≈ 1.8 (Pure DNA) vs. 0.6 (Pure Protein) — Purine/pyrimidine bases absorb strongly at 260 nm, while Trp/Tyr side chains absorb at 280 nm.

Pulse Oximetry (Oxyhemoglobin vs. Deoxyhemoglobin) — O_2 Binding Shifts Heme Porphyrin Absorption (660 nm vs 940 nm) — Changes in iron d-orbital coordination alter the porphyrin π→π* transition, turning venous blood dark red and arterial blood bright crimson.

Screenshot: uv-visuals



uv:step-1 | 1. Follow conjugation

Alternating C=C–C=C bonds create a connected p-orbital system. Adjacent p orbitals must remain sufficiently aligned for conjugation.

Displayed molecule ID: butadiene

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Screenshot: uv-step-1



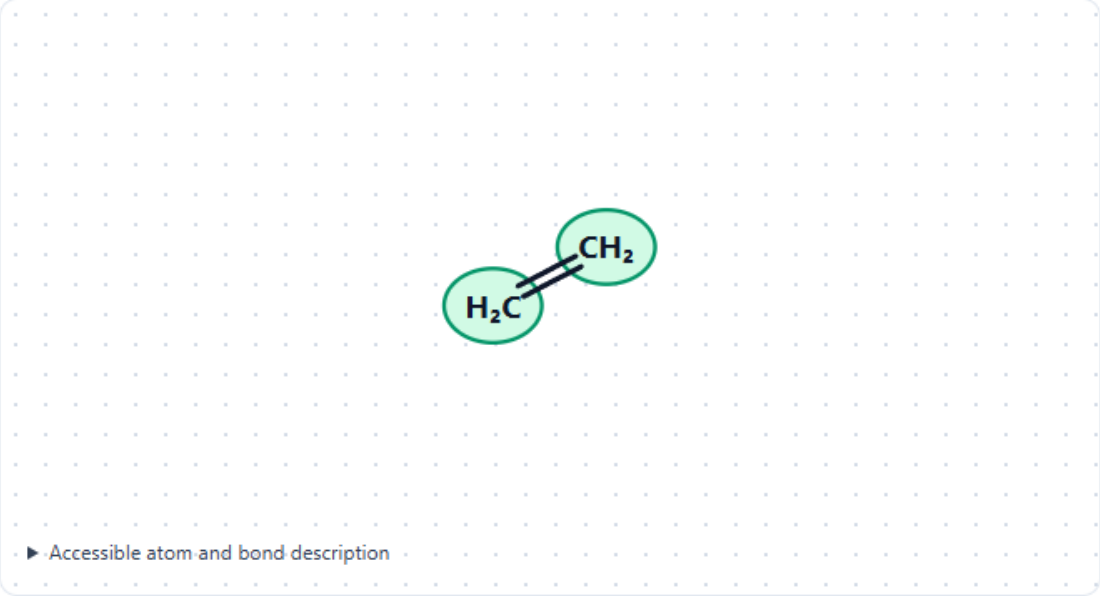
uv:step-2 | 2. Connect wavelength to energy

Relative to an isolated comparable alkene, conjugation often reduces the pi-to-pi-star transition energy. Use E=hc/λ to translate that smaller gap into longer wavelength.

Displayed molecule ID: ethene

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Screenshot: uv-step-2



nmr | Proton NMR: shift, integration, splitting

Three independent kinds of evidence

nmr:rule | Core Rules

Chemical shift reports environment; integration reports relative proton count; splitting reports coupling. In simple first-order cases, coupling to n equivalent neighboring protons produces n+1 lines. Equivalent protons do not split one another. Deshielded protons appear downfield at higher ppm.

Visual card: ^{1}H and ^{13}C NMR: Symmetry Equivalence, Chemical Shift (δ ppm), Integration & (n + 1) Splitting

Chemistry diagram: Matched molecule and illustrative NMR signals. Peak height is not integral; use the signal labels and complete pattern.

Signals = Symmetry Sets

Shift δ (ppm): Downfield = Deshielded (Left)

Integration = H Ratio

Splitting = n + 1 Neighbors

Aldehyde (RCHO) / COOH | δ 9–10 ppm (Ald) | δ 10–13 ppm (COOH) | Extreme deshielding (C=O anisotropy + induction)

Aromatic (Ar–H) / Alkene | δ 6.5–8.5 ppm (Ar) | δ 4.5–6.5 ppm (C=C) | π ring current / anisotropy deshields protons

α to O/N/X (–OCH_2–) | δ 3.0–4.5 ppm | Electronegative heteroatom pulls e^- density

Alkyl / α to C=O | δ 0.8–1.8 ppm (R–H) | δ 2.0–2.5 ppm (α-C=O) | Shielded upfield near TMS reference (0.0 ppm)

Number of Signals = Chemically Non-Equivalent Proton Sets — — Protons interchangeable by a rotational axis (C_n) or mirror plane (homotopic/enantiotopic in achiral solvent) give ONE shared signal.

Chemical Shift (δ ppm from TMS = 0 ppm) — — Downfield (higher ppm, to the LEFT) = DESHIELDED by electronegative atoms (O, N, Halogens) or π-bond magnetic anisotropy (Aldehyde 9–10, Aromatic 7–8, Alkene 5–6, O–CH 3.5–4, α-C=O 2.1, Alkyl 0.9–1.5).

Integration (Area = Proton Ratio) & Spin–Spin Splitting (n + 1 Rule) — — Peak area ratio gives the relative number of H atoms in each set; n non-equivalent neighboring H atoms (within 3 bonds, H–C–C–H) split a signal into (n + 1) peaks (Pascal's triangle: 1:1 doublet, 1:2:1 triplet, 1:3:3:1 quartet).

nmr:cue | Memory Cues

Shift = neighborhood; area = population; splitting = neighbors. Upfield means lower ppm and greater shielding; downfield means higher ppm. Tetramethylsilane (TMS) is commonly assigned δ = 0 ppm. Coupling constant J is the line spacing in Hz, not the integral or chemical shift.

Visual card: Classic NMR Splitting Patterns & Exchangeable Proton Tricks

Chemistry diagram: Illustrative integrated signal areas for the same graph; line height is not proton count.

What NMR pattern do you observe in the spectrum?

3H Triplet (~1.2 ppm) + 2H Quartet (~3.5–4.1 ppm) — Unmistakable ETHYL Group (–CH_2CH_3)! — The CH_3 (3H) is split by 2 neighbors into a triplet (2+1=3); the CH_2 (2H) is split by 3 neighbors into a quartet (3+1=4).

6H Doublet (~1.1 ppm) + 1H Septet (~2.5–4.0 ppm) — Unmistakable ISOPROPYL Group (–$CH(CH_3)_2$)! — Two symmetric methyls (6H) are split by 1 methine H into a doublet; the single CH (1H) is split by 6 methyl H's into a 7-line septet.

1H Broad Singlet That Vanishes After Adding D_2O — Alcohol (–OH) or Amine (–NH) Proton! — Rapid intermolecular proton exchange averages out spin-spin coupling (usually a broad singlet) and swaps –OH for NMR-silent –OD in D_2O!

nmr:trap | Common Mistakes / MCAT Traps

n+1 is a limited first-order rule, not universal. Nonequivalent neighbors can give more complex patterns; exchangeable OH/NH protons often do not show the expected simple coupling. Chemical equivalence depends on symmetry and timescale. NMR requires nonzero nuclear spin. Odd-mass isotopes such as ^{1}H and ^{13}C are active; odd-atomic-number nuclei with even mass, such as ^{2}H and ^{14}N, can also be active. Even-proton/even-neutron nuclei such as ^{12}C and ^{16}O have spin zero. Some active isotopes have broad or difficult signals, so activity alone does not guarantee easy detection.

Visual card: Using Peak Height Instead of Integral Area or Splitting by Equivalent / Same-Carbon Protons

Common Trap — — Saying a tall singlet has more protons than a wide multiplet based on peak height, or splitting a CH_3 group into a quartet using its own 3 protons.

Correct Reasoning — — Integral measures signal area, not height. The simple n+1 rule assumes first-order coupling to equivalent neighbors; magnetic equivalence and coupling patterns require more care in complex systems.

nmr:app | MCAT Applications

Typical rough proton regions: alkyl 0–3 ppm; C next to O often 3–4.5; vinylic about 4.5–6.5; aromatic 6–8.5; aldehydic 9–10; acid OH often 10–13. Use the full pattern.

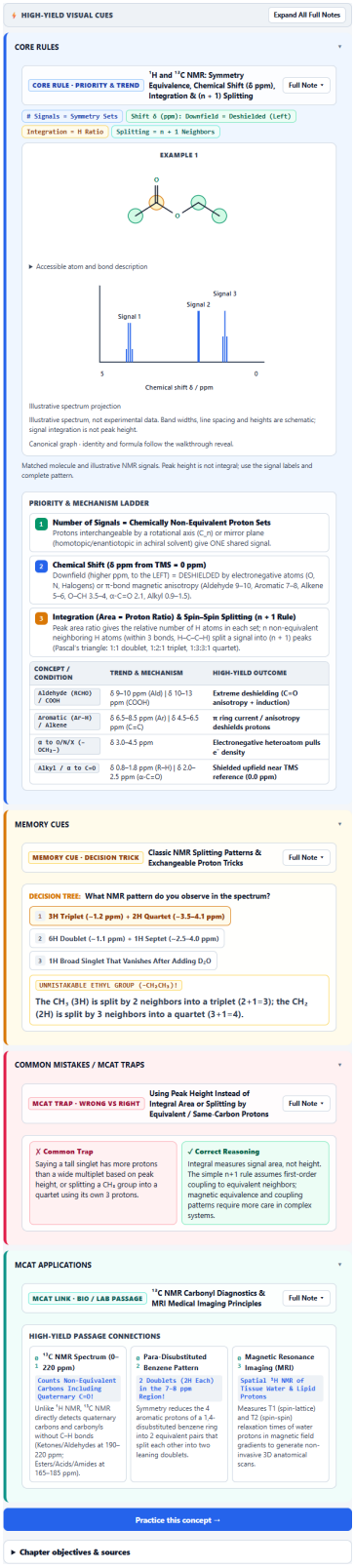
Visual card: ^{13}C NMR Carbonyl Diagnostics & MRI Medical Imaging Principles

^{13}C NMR Spectrum (0–220 ppm) — Counts Non-Equivalent Carbons Including Quaternary C=O! — Unlike ^{1}H NMR, ^{13}C NMR directly detects quaternary carbons and carbonyls without C–H bonds (Ketones/Aldehydes at 190–220 ppm; Esters/Acids/Amides at 165–185 ppm).

Para-Disubstituted Benzene Pattern — 2 Doublets (2H Each) in the 7–8 ppm Region! — Symmetry reduces the 4 aromatic protons of a 1,4-disubstituted benzene ring into 2 equivalent pairs that split each other into two leaning doublets.

Magnetic Resonance Imaging (MRI) — Spatial ^{1}H NMR of Tissue Water & Lipid Protons — Measures T1 (spin-lattice) and T2 (spin-spin) relaxation times of water protons in magnetic field gradients to generate non-invasive 3D anatomical scans.

Screenshot: nmr-visuals



nmr:step-1 | 1. Three proton sets

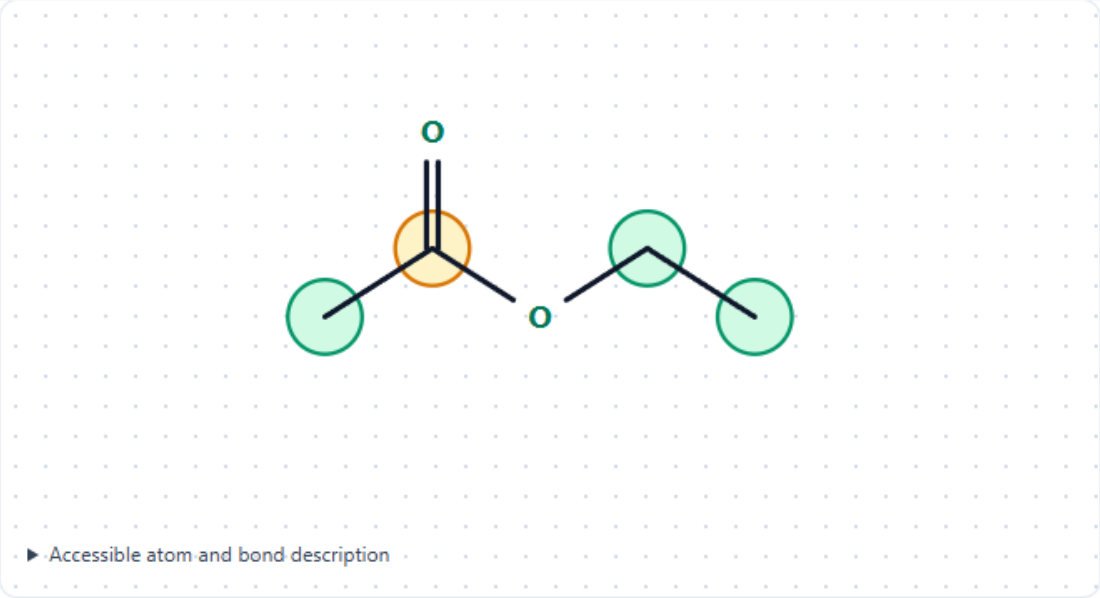
Ethyl ethanoate has acyl methyl, OCH2, and terminal ethyl methyl proton sets. Their integrals are 3:2:3. The carbonyl carbon has no H. The three green carbon sites carry separate proton environments, not one combined set. Their positions and

integrals distinguish them; the orange carbonyl carbon carries no H.

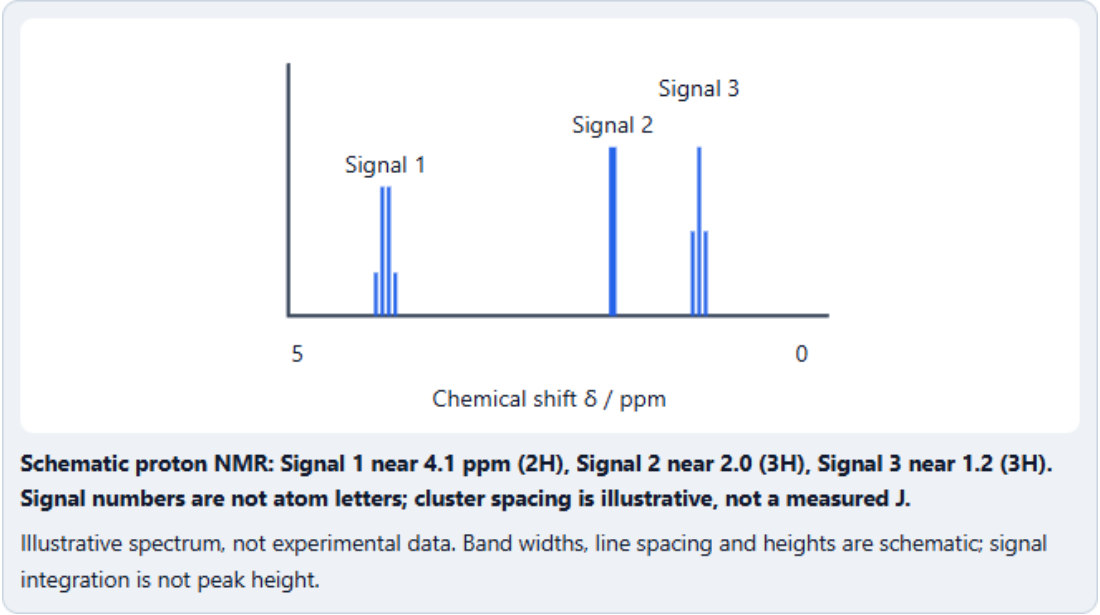
Displayed molecule ID: ethylacetate

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Screenshot: nmr-step-1



Screenshot: nmr-step-1-diagram



Schematic proton NMR: Signal 1 near 4.1 ppm (2H), Signal 2 near 2.0 (3H), Signal 3 near 1.2 (3H). Signal numbers are not atom letters; cluster spacing is illustrative, not a measured J.

Illustrative spectrum, not experimental data. Band widths, line spacing and heights are schematic; signal integration is not peak height.

{"diagramSpecification": {"type": "spectrum", "min": 0, "max": 5, "unit": "Chemical shift δ / ppm", "caption": "Schematic proton NMR: Signal 1 near 4.1 ppm (2H), Signal 2 near 2.0 (3H), Signal 3 near 1.2 (3H). Signal numbers are not atom letters; cluster spacing is illustrative, not a measured J.", " peaks": [{"x": 4.1, "height": 0.65, "label": "Signal 1", "lines": 4, "spacing": 0.06}, {"x": 2, "height": 0.85, "label": "Signal 2", "lines": 1}, {"x": 1.2, "height": 0.85, "label": "Signal 3", "lines": 3, "spacing": 0.06}]}}

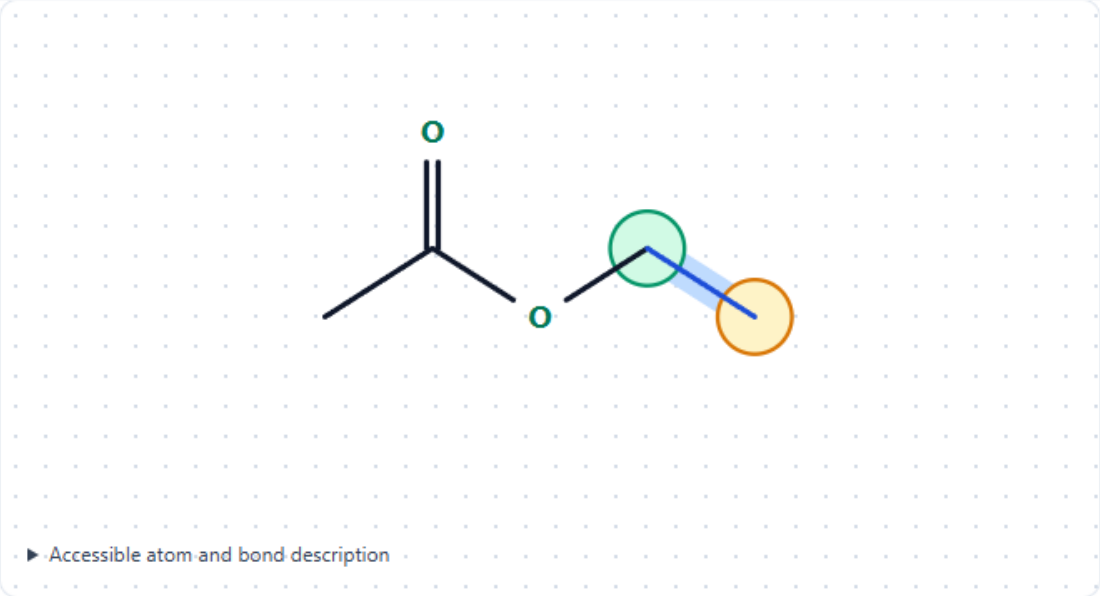
nmr:step-2 | 2. Ethyl splitting

OCH2 couples to three neighboring methyl H atoms, giving a quartet in a simple first-order picture. The terminal CH3 couples to two methylene H atoms, giving a triplet.

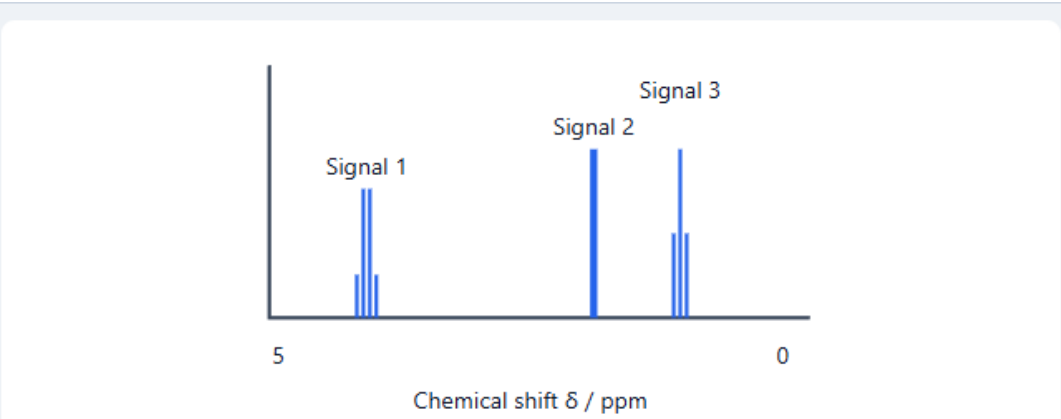
Displayed molecule ID: ethylacetate

{"highlights": {"path": [4, 5], "prin": [4], "sub": [5]}, "revealName": false}

Screenshot: nmr-step-2



Screenshot: nmr-step-2-diagram



Schematic proton NMR: Signal 1 near 4.1 ppm (2H), Signal 2 near 2.0 (3H), Signal 3 near 1.2 (3H). Signal numbers are not atom letters; cluster spacing is illustrative, not a measured J.

Illustrative spectrum, not experimental data. Band widths, line spacing and heights are schematic; signal integration is not peak height.

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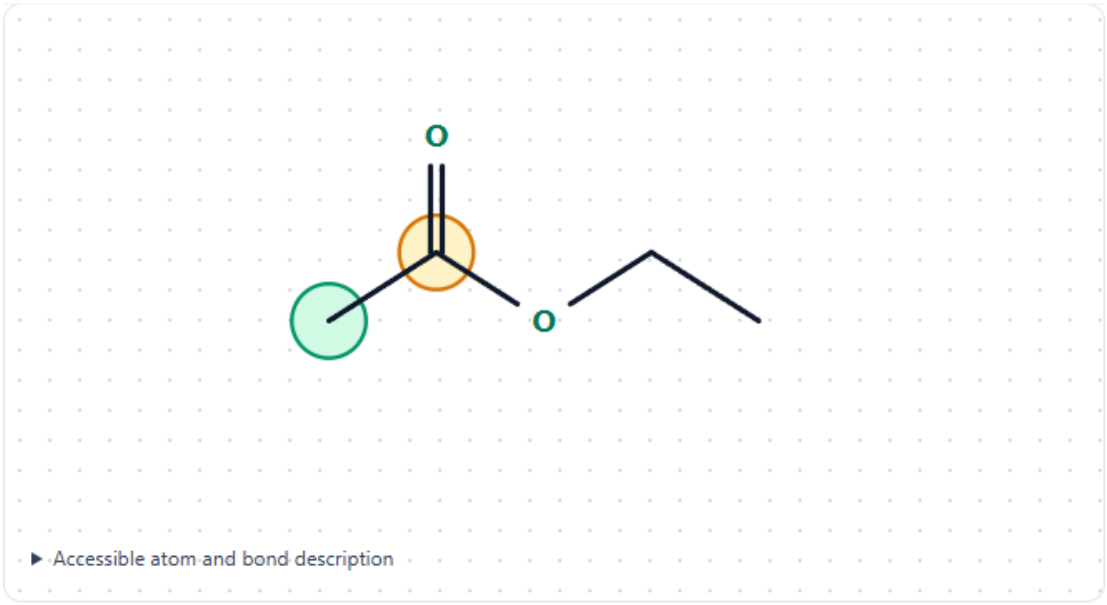
nmr:step-3 | 3. Acyl methyl

The acyl methyl has no H-bearing carbon immediately across the carbonyl bond, so it is approximately a singlet in this example. Do not apply n+1 by counting every H anywhere in the molecule.

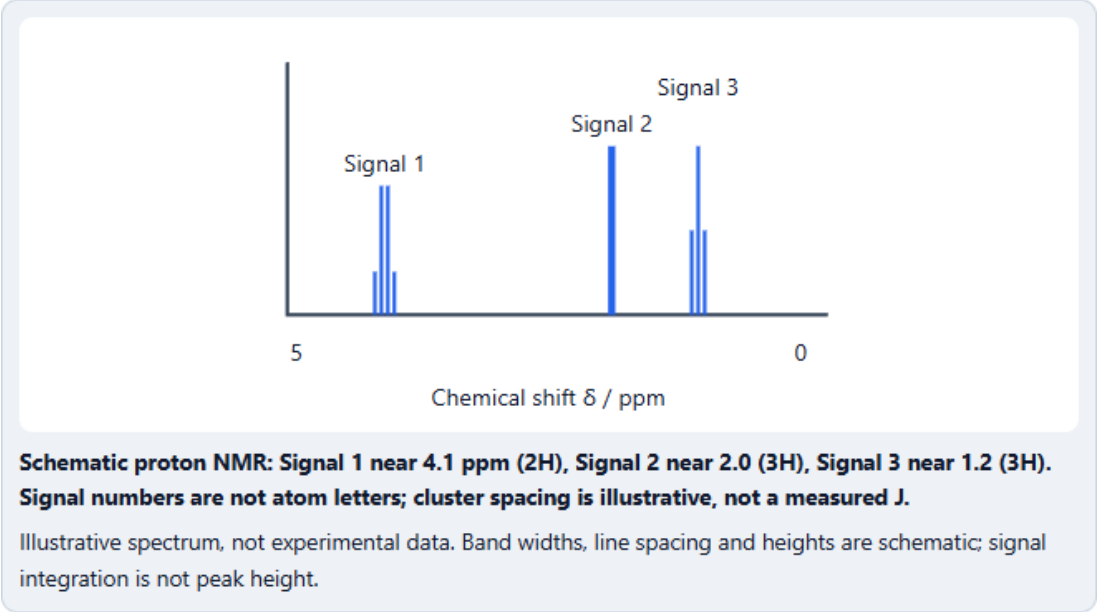
Displayed molecule ID: ethylacetate

{"highlights": {"prin": [0], "sub": [1]}, "revealName": false}

Screenshot: nmr-step-3



Screenshot: nmr-step-3-diagram



{"diagramSpecification": {"type": "spectrum", "min": 0, "max": 5, "unit": "Chemical shift δ / ppm", "caption": "Schematic proton NMR: Signal 1 near 4.1 ppm (2H), Signal 2 near 2.0 (3H), Signal 3 near 1.2 (3H). Signal numbers are not atom letters; cluster spacing is illustrative, not a measured J.", " peaks": [{"x": 4.1, "height": 0.65, "label": "Signal 1", "lines": 4, "spacing": 0.06}, {"x": 2, "height": 0.85, "label": "Signal 2", "lines": 1}, {"x": 1.2, "height": 0.85, "label": "Signal 3", "lines": 3, "spacing": 0.06}]}}

combine | Combine formula, IR, and NMR

Independent clues should agree

combine:rule | Core Rules

For ordinary neutral closed-shell organic formulas, degree of unsaturation (DBE) = (2C+2+N−H−X)/2; O and divalent S do not enter. A ring or double bond contributes one DBE; a triple bond contributes two. Then combine IR group evidence with NMR environments and integrations.

Visual card: Integrated Structure Elucidation Workflow: Formula/MS → IHD → IR → $^{1}H/^{13}C$ NMR

Chemistry diagram: Same formula, different connectivity: compare the two carbonyl isomers.

1. Formula → IHD

2. IR → Functional Group (C=O, O–H)

3. NMR Symmetry + Shifts + Splitting

4. MS: M^{+}, Br (1:1), Cl (3:1)

Step 1: Formula & MS | IHD = (2C+2+N−H−X)/2; Odd MW = Odd # of N | M+2 peak: Br = 1:1 ($^{79}Br/^{81}Br$); Cl = 3:1 ($^{35}Cl/^{37}Cl$)

Step 2: IR Spectrum | 1715 cm^{-1} = C=O; 3350 cm^{-1} = O–H | Immediately tells you where the oxygen/nitrogen lives

Step 3: ^{1}H & ^{13}C NMR | Count signals (symmetry) + integration + (n+1) | Snap together fragments (ethyl, isopropyl, para-aryl)

Step 1: Molecular Formula, Nitrogen Rule & IHD — — Calculate IHD = (2C + 2 + N − H − X)/2. If IHD ≥ 4, suspect a benzene ring (4 IHD). Check Mass Spec for M^{+} molecular ion and halogen M+2 isotopes (Cl = 3:1; Br = 1:1).

Step 2: Pinpoint Functional Groups by IR — — Check 1650–1800 cm^{-1} for C=O and 2500–3600 cm^{-1} for O–H / N–H. Combine IHD + IR (e.g., 1 IHD + 1715 cm^{-1} IR = ketone or aldehyde; 0 IHD + 3350 cm^{-1} = saturated alcohol).

Step 3: Assemble Fragments with ^{1}H & ^{13}C NMR — — Use signal count to spot symmetry, integration to distribute H counts, chemical shift (δ) to place groups next to C=O or O, and (n + 1) splitting to connect adjacent CH_x neighbors.

combine:cue | Memory Cues

Formula constrains possibilities; IR finds groups; NMR organizes the skeleton.

Visual card: Distinguishing Constitutional Isomers Using Combined IR + NMR Clues

How do you quickly distinguish common isomer pairs on the MCAT?

3-Pentanone vs. 2-Pentanone (Both $C_5H_{10}O$, 1715 cm^{-1}) — 3-Pentanone Has Symmetry → Only 2 ^{1}H Signals (6H Triplet + 4H Quartet) And 3 ^{13}C Signals! — 2-Pentanone is unsymmetrical (4 ^{1}H signals including a 3H methyl ketone singlet at 2.1 ppm, plus 5 ^{13}C signals).

Ethyl Acetate ($CH_3COOCH_2CH_3$) vs. Methyl Propanoate ($CH_3CH_2COOCH_3$) — Check Which Group Is at ~4.1 ppm (Bonded to –O–)! — In ethyl acetate, the 2H QUARTET is at 4.1 ppm ($–OCH_2CH_3$) and the 3H singlet is at 2.0 ppm; in methyl propanoate, the 3H SINGLET is at 3.7 ppm ($–OCH_3$)!

Aldehyde vs. Ketone Isomers — Aldehyde Shows a 1H Peak at 9.5–10 ppm NMR + 2720/2820 cm^{-1} IR! — A ketone lacks the downfield 9.5 ppm aldehydic proton and the C–H Fermi doublet in IR.

combine:trap | Common Mistakes / MCAT Traps

A single NMR signal can represent several equivalent groups. DBE says how much unsaturation exists, not where it is. Do not force phosphorus-rich or unusual-valence formulas into this simple equation.

Visual card: Confusing the Base Peak (Tallest MS Peak = 100%) with the Parent Molecular Ion (M^{+}•)

Common Trap — — Assuming the tallest peak in a mass spectrum is always the molecular weight (M^+), or ignoring symmetry when counting ^{13}C NMR peaks.

Correct Reasoning — — The BASE PEAK is simply the most stable carbocation fragment (100% relative abundance, e.g., tropylium m/z 91 or acylium RCO^+); the parent ion $M^{+}\bullet$ is the highest m/z peak (excluding M+1/M+2 isotopes)!

combine:app | MCAT Applications

Passage-based structure identification rewards eliminating structures inconsistent with even one reliable observation.

Visual card: Unknown Metabolite Identification & Isotope Ratio Mass Spectrometry

Bromine vs. Chlorine MS Isotope Signature — M : M+2 = 1:1 (Bromine $^{79}Br/^{81}Br$) vs. 3:1 (Chlorine $^{35}Cl/^{37}Cl$) — Twin parent peaks of equal height separated by 2 m/z units instantly prove the molecule contains one bromine atom.

Nitrogen Rule in Mass Spectrometry — Odd M^+ Molecular Weight = Odd Number of Nitrogens (1, 3, 5...) — Because N is trivalent with an even atomic mass (14), a neutral organic molecule with 1 nitrogen always has an ODD molecular weight!

Benzylic Fragmentation (m/z = 91) — Alkylbenzenes Cleave at the Benzylic Bond → $C_7H_7^+$ (m/z 91) — Forms the exceptionally stable aromatic tropylium cation (6π electrons in a 7-membered ring) as the base peak.

Screenshot: combine-visuals



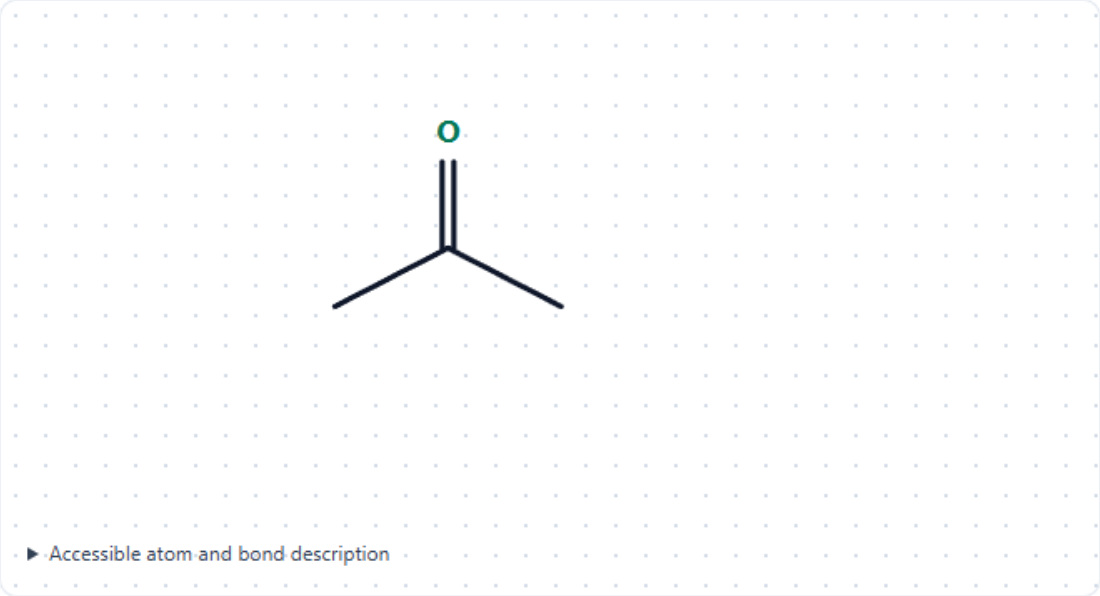
combine:step-1 | 1. Formula constraint

C3H6O has DBE=(2×3+2−6)/2=1. A carbonyl can account for that entire unit of unsaturation.

Displayed molecule ID: acetone

{"highlights": {}, "revealName": false}

Screenshot: combine-step-1



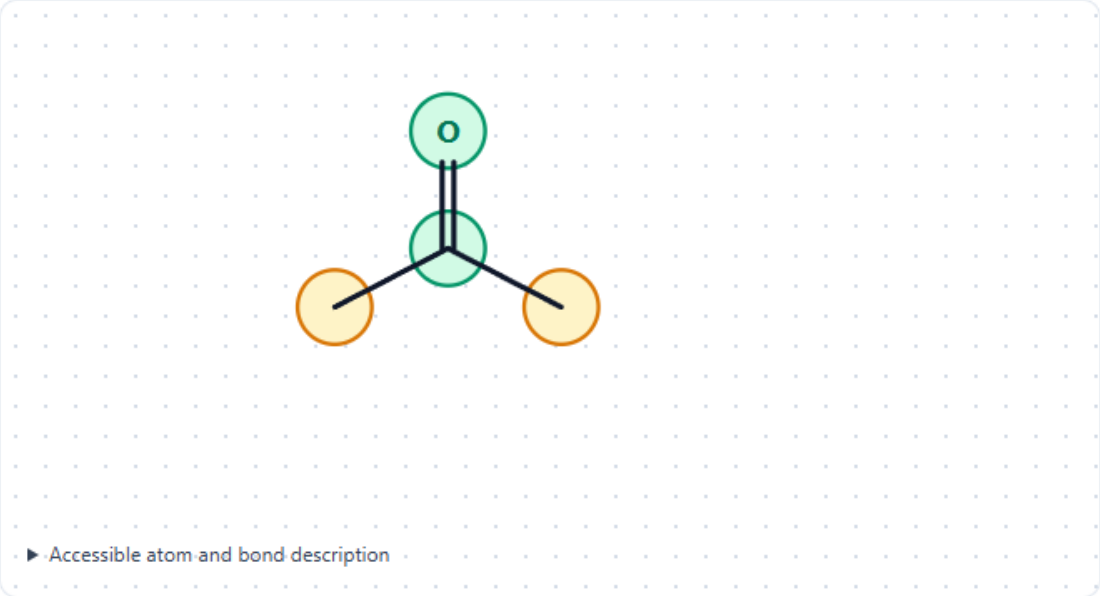
combine:step-2 | 2. Add two measurements

Suppose IR shows a ketone-like C=O band and proton NMR shows a single 6H singlet near 2 ppm. Two symmetry-equivalent methyl groups adjacent to C=O fit those data.

Displayed molecule ID: acetone

{"highlights": {"prin": [1, 3], "sub": [0, 2]}, "revealName": true}

Screenshot: combine-step-2



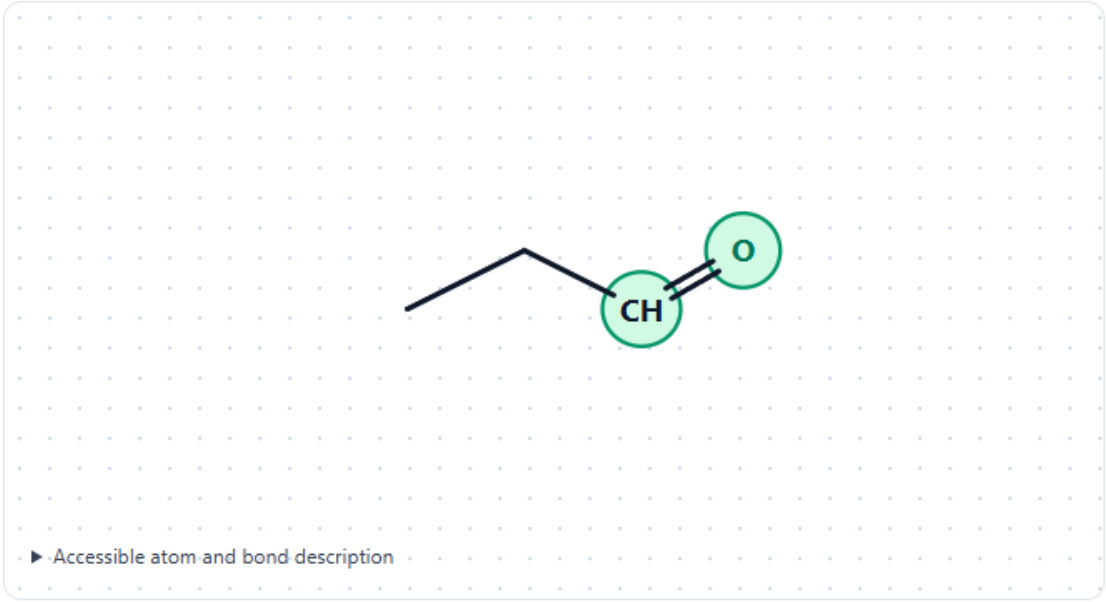
combine:step-3 | 3. Challenge an alternative

Propanal shares C3H6O and DBE=1 with acetone, and both have C=O absorption. But propanal has an aldehydic H (shown explicitly as CH at the terminal carbonyl carbon) and multiple proton environments; it does not fit the single 6H singlet. Compare every observation, not just formula and IR.

Displayed molecule ID: propanal

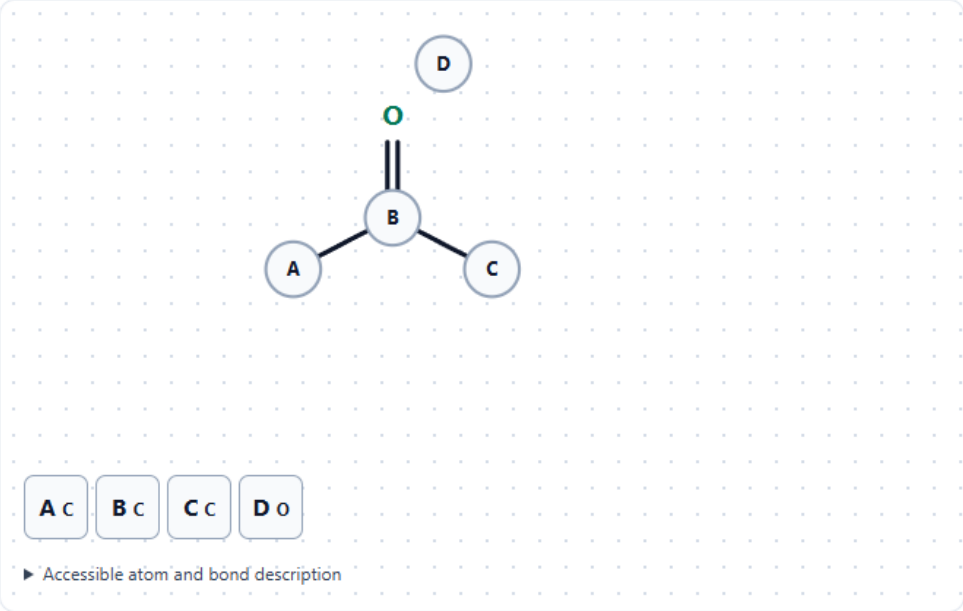
{"highlights": {"prin": [2, 3]}, "revealName": true}

Screenshot: combine-step-3

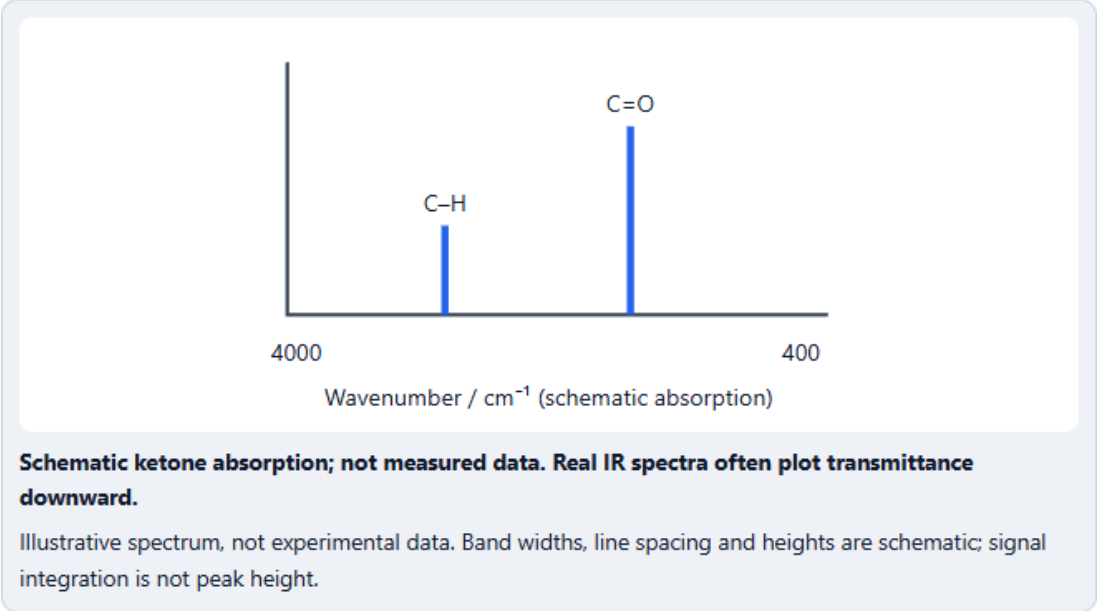


ir-read | Interpret functional-group evidence

Screenshot: ir-read-stage-1-molecule



Screenshot: ir-read-diagram

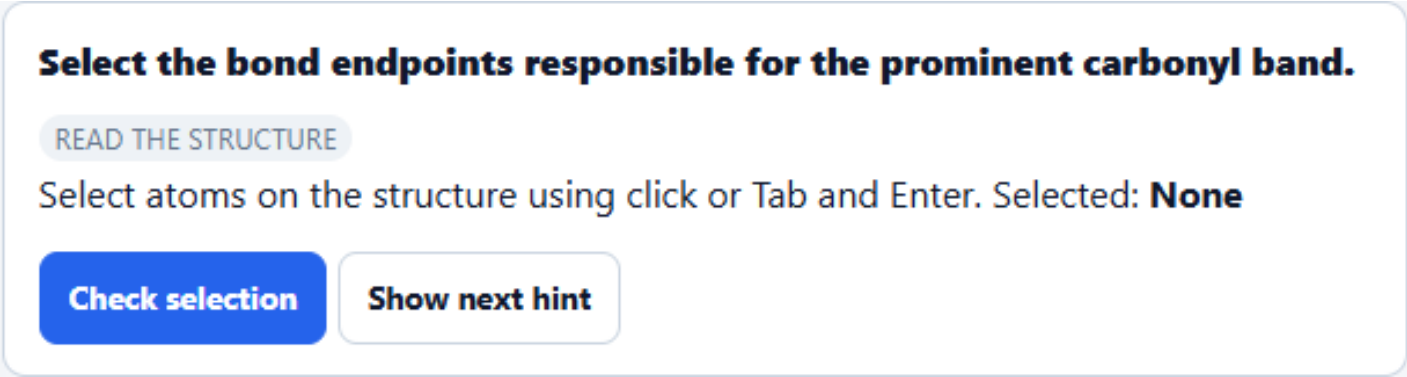


{"diagramSpecification": {"type": "spectrum", "min": 400, "max": 4000, "unit": "Wavenumber / cm^{-1} (schematic absorption)", "caption": "Schematic ketone absorption; not measured data. Real IR spectra often plot transmittance downward.", "peaks": [{"x": 1715, "height": 0.95, "label": "C=O"}, {"x": 2950, "height": 0.45, "label": "C–H"}]}}

ir-read:stage-1 | Select the bond endpoints responsible for the prominent carbonyl band.

Answer type: atoms

Screenshot: ir-read-stage-1-question



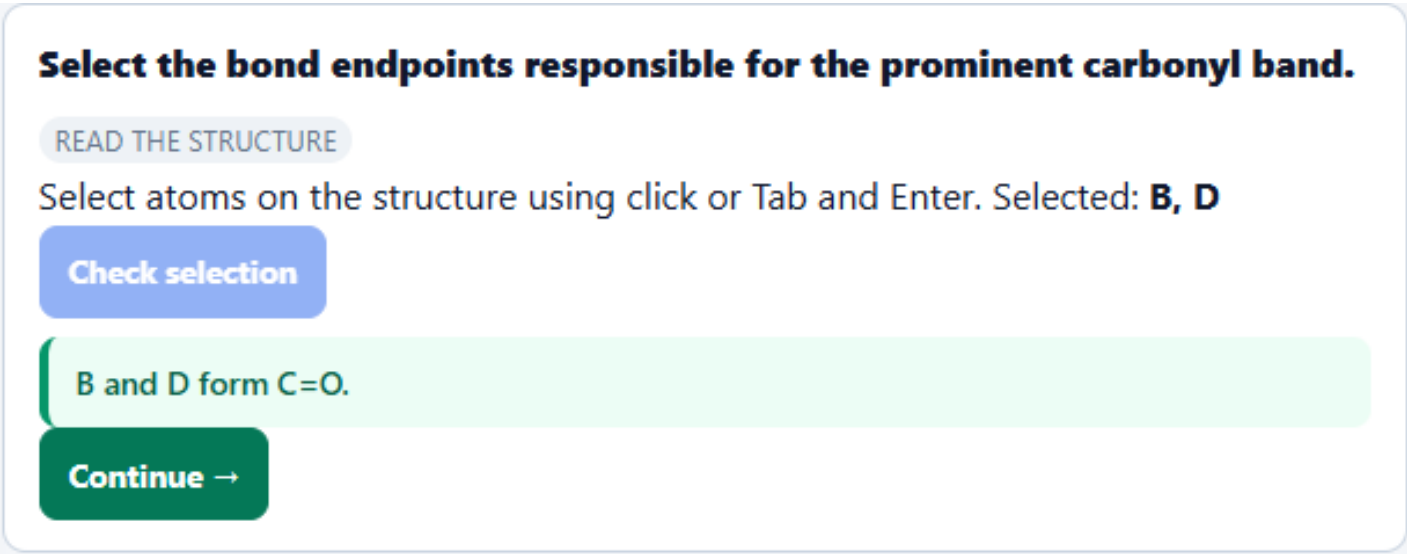
{"authorAnswer": [1, 3], "tolerance": null, "highlightsAfterCorrectAnswer": {"prin": [1, 3]}}

Hint 1: Find the double bond to O.

Hint 2: Select both carbon and oxygen.

Explanation: B and D form C=O.

Screenshot: ir-read-stage-1-feedback



ir-read:stage-2 | An IR-active vibration must change...

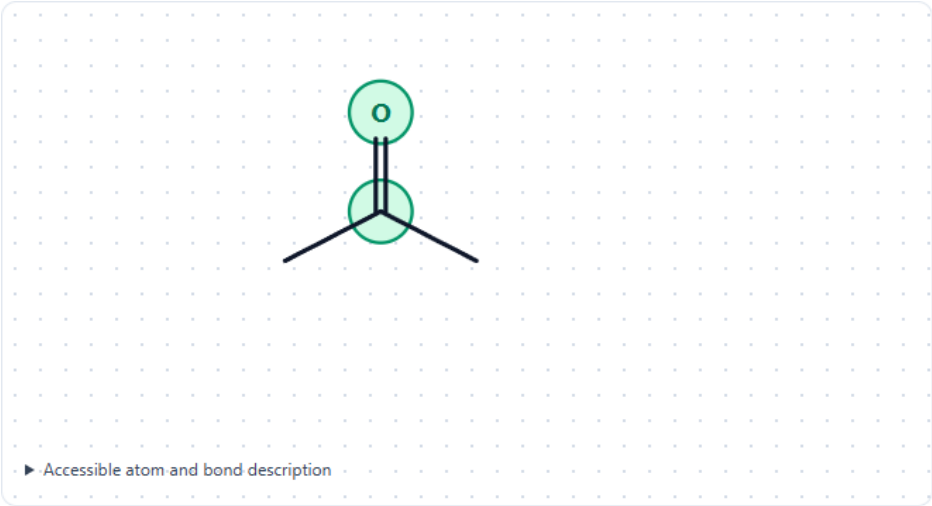
Answer type: choice

Option 0 (A): Dipole moment

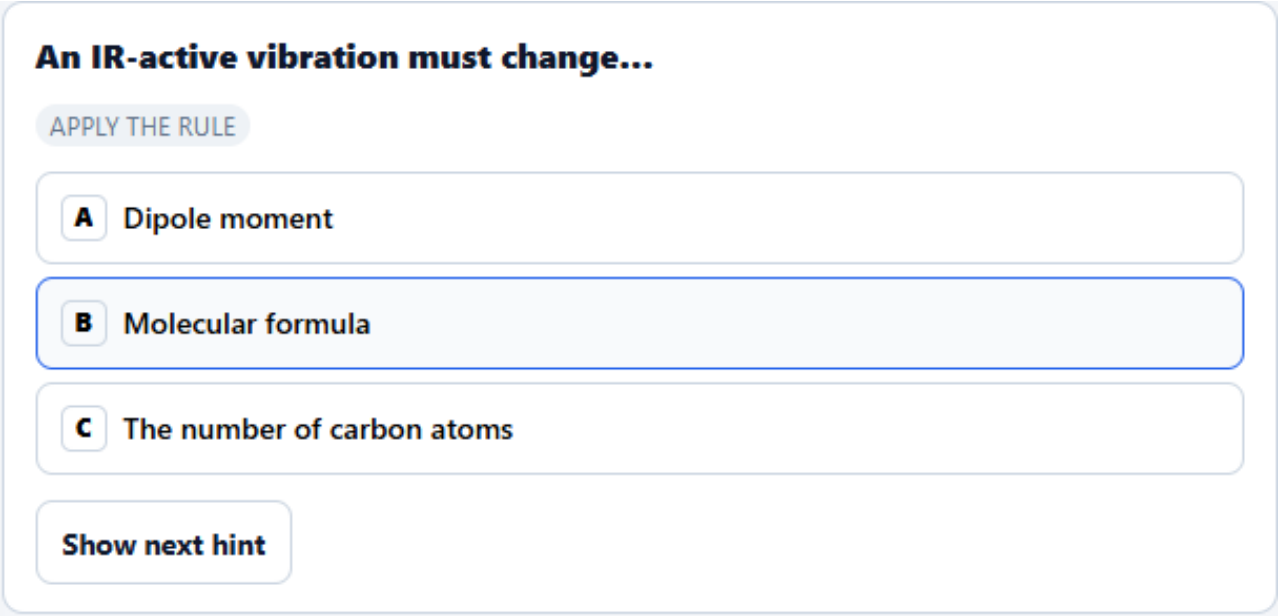
Option 1 (B): Molecular formula

Option 2 (C): The number of carbon atoms

Screenshot: ir-read-stage-2-molecule



Screenshot: ir-read-stage-2-question



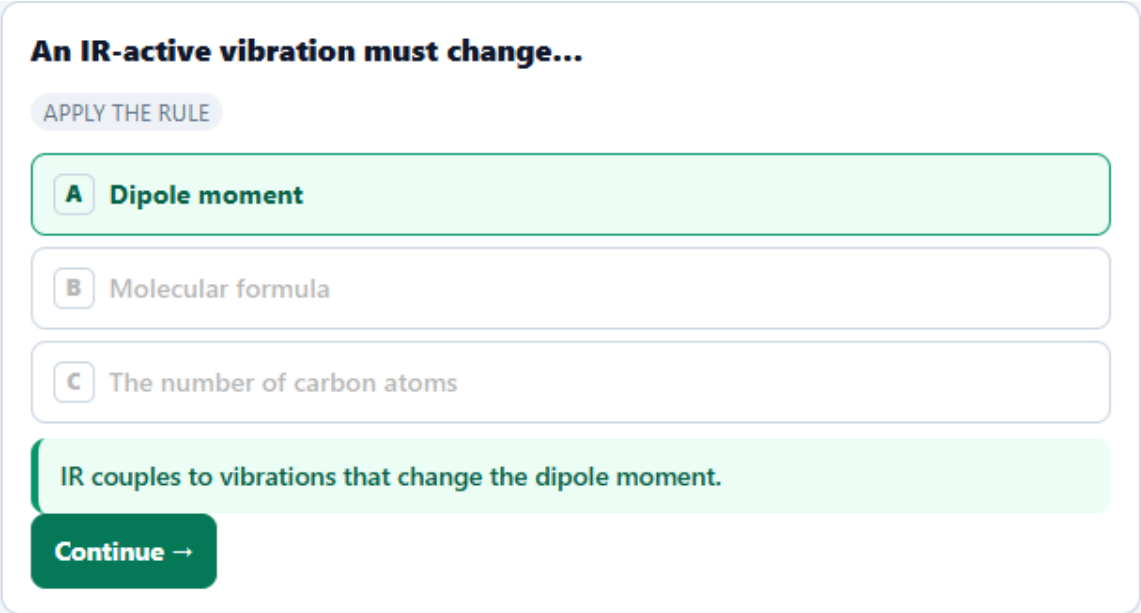
{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Think about interaction with oscillating electric fields.

Hint 2: Not every possible vibration absorbs IR.

Explanation: IR couples to vibrations that change the dipole moment.

Screenshot: ir-read-stage-2-feedback



This stage uses a different molecule (connectivity shown).

ir-read:stage-3 | A broad 3200–3600 cm^{-1} band appearing after reduction supports...

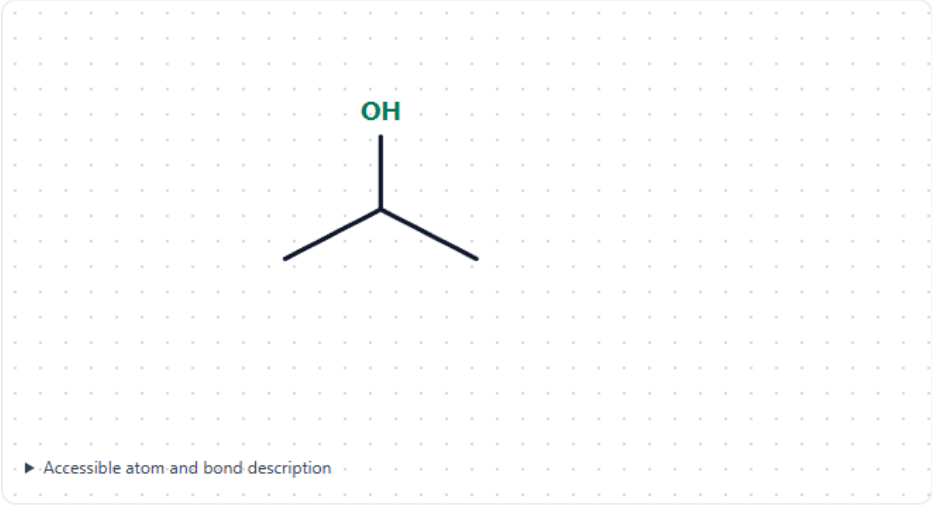
Answer type: choice

Option 0 (A): An alcohol OH

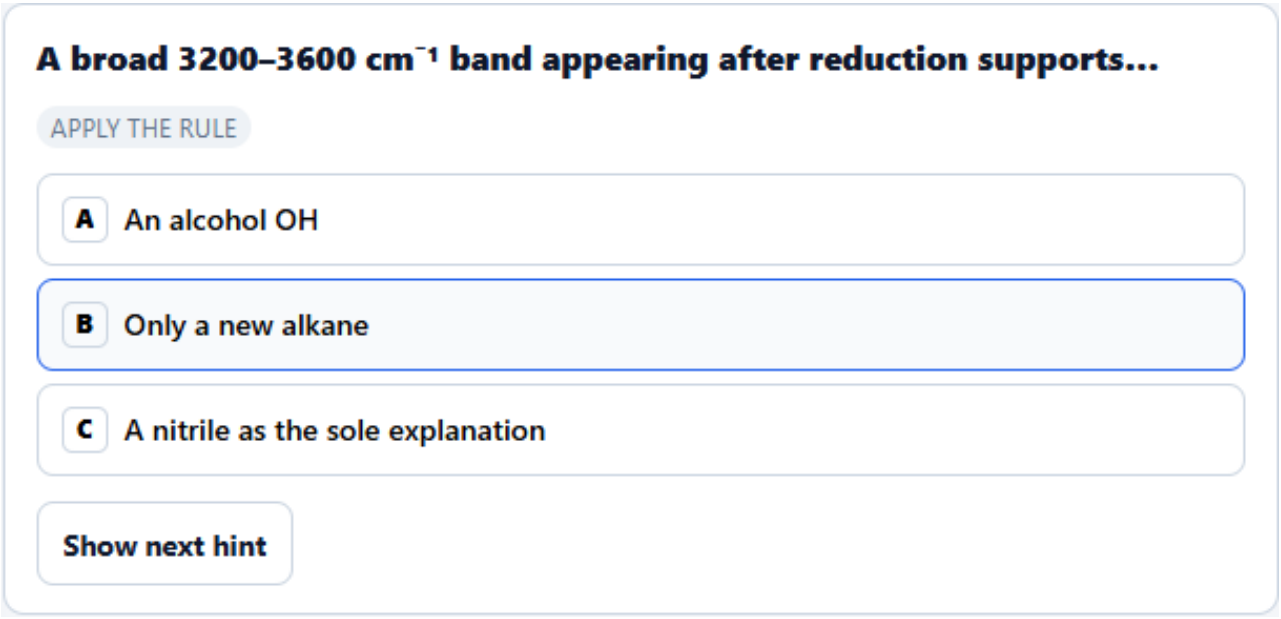
Option 1 (B): Only a new alkane

Option 2 (C): A nitrile as the sole explanation

Screenshot: ir-read-stage-3-molecule



Screenshot: ir-read-stage-3-question



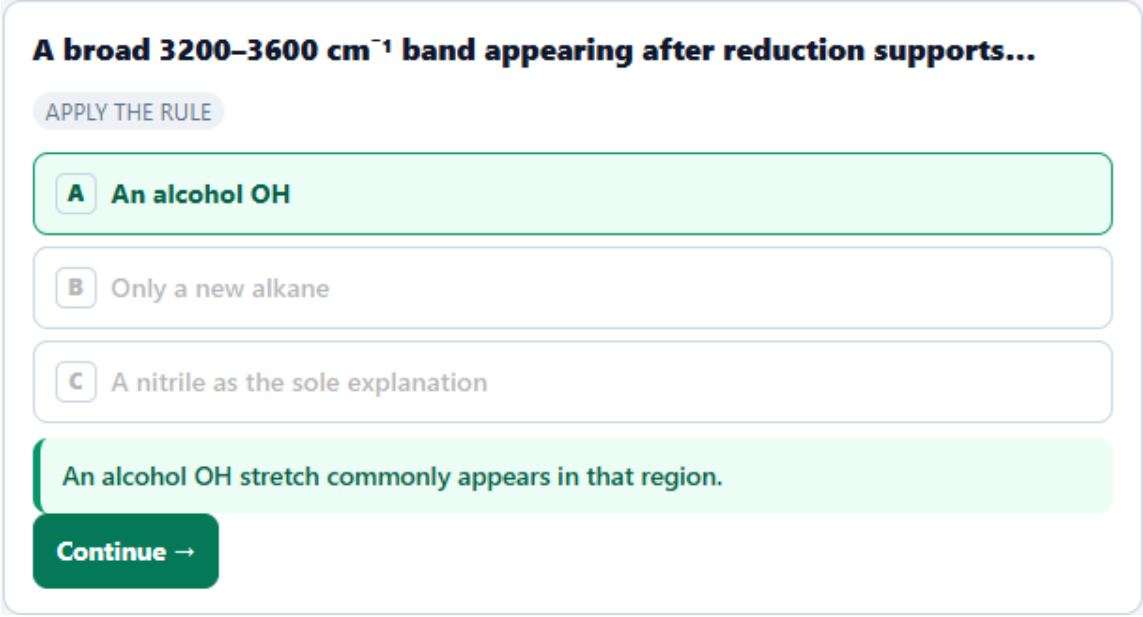
{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Reduction retains oxygen.

Hint 2: C=O becomes C–OH.

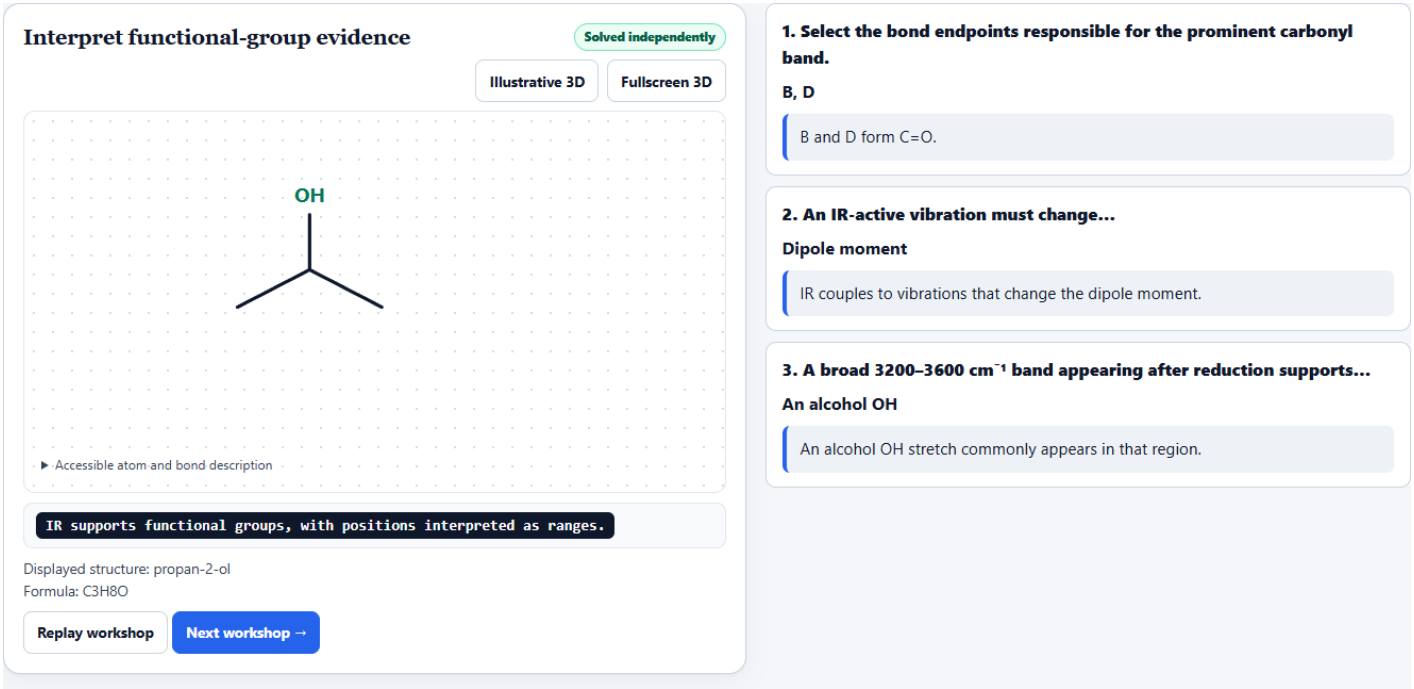
Explanation: An alcohol OH stretch commonly appears in that region.

Screenshot: ir-read-stage-3-feedback



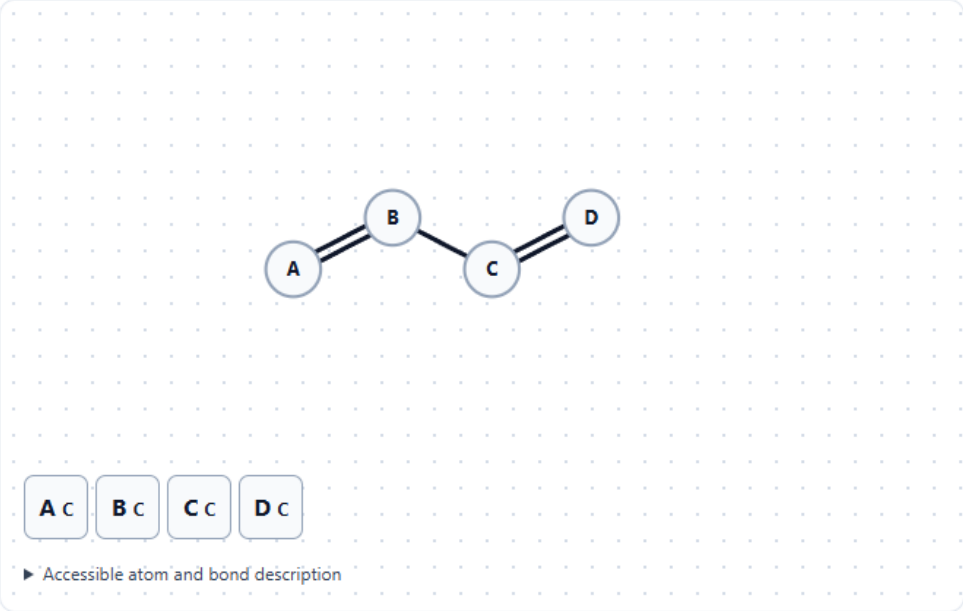
Completion summary: IR supports functional groups, with positions interpreted as ranges.

Screenshot: ir-read-complete



uv-energy | Connect conjugation and absorbance

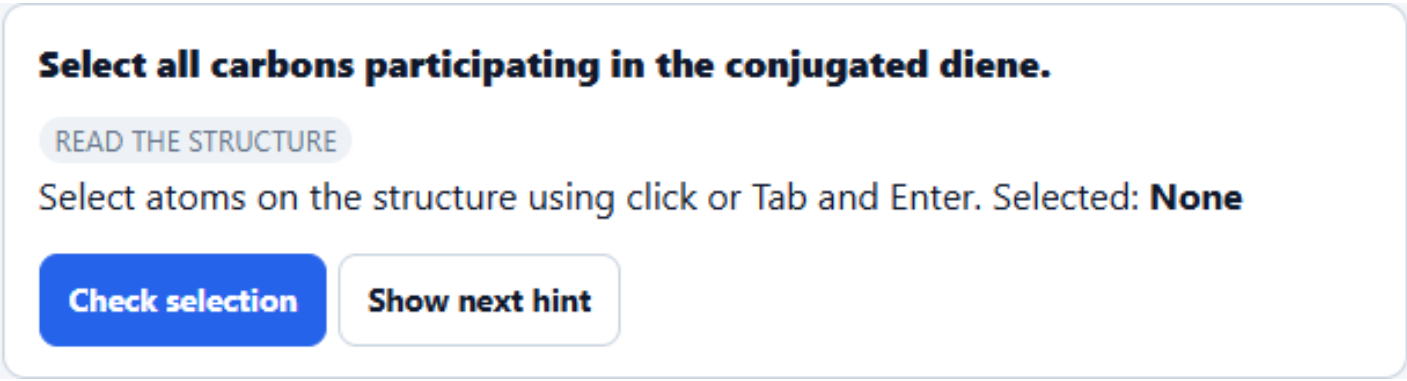
Screenshot: uv-energy-stage-1-molecule



uv-energy:stage-1 | Select all carbons participating in the conjugated diene.

Answer type: atoms

Screenshot: uv-energy-stage-1-question



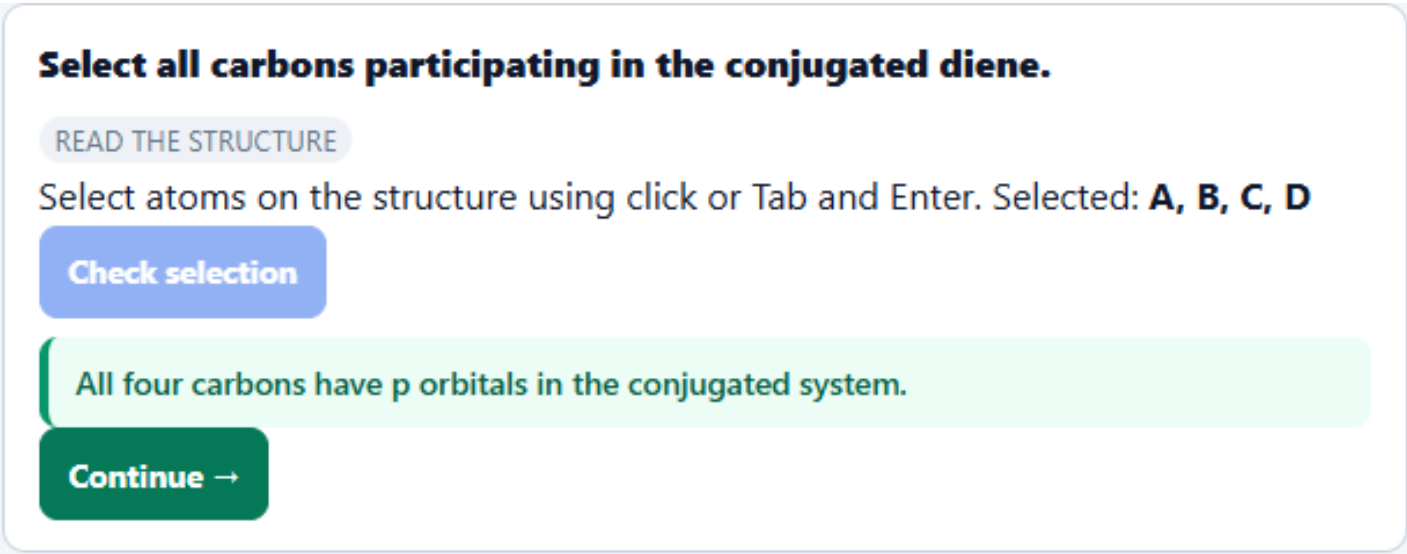
{"authorAnswer": [0, 1, 2, 3], "tolerance": null, "highlightsAfterCorrectAnswer": {"path": [0, 1, 2, 3]}}

Hint 1: Follow the alternating double–single–double pattern.

Hint 2: Include the carbons of both double bonds.

Explanation: All four carbons have p orbitals in the conjugated system.

Screenshot: uv-energy-stage-1-feedback



uv-energy:stage-2 | For comparable pi systems, increased conjugation often shifts absorption to...

Answer type: choice

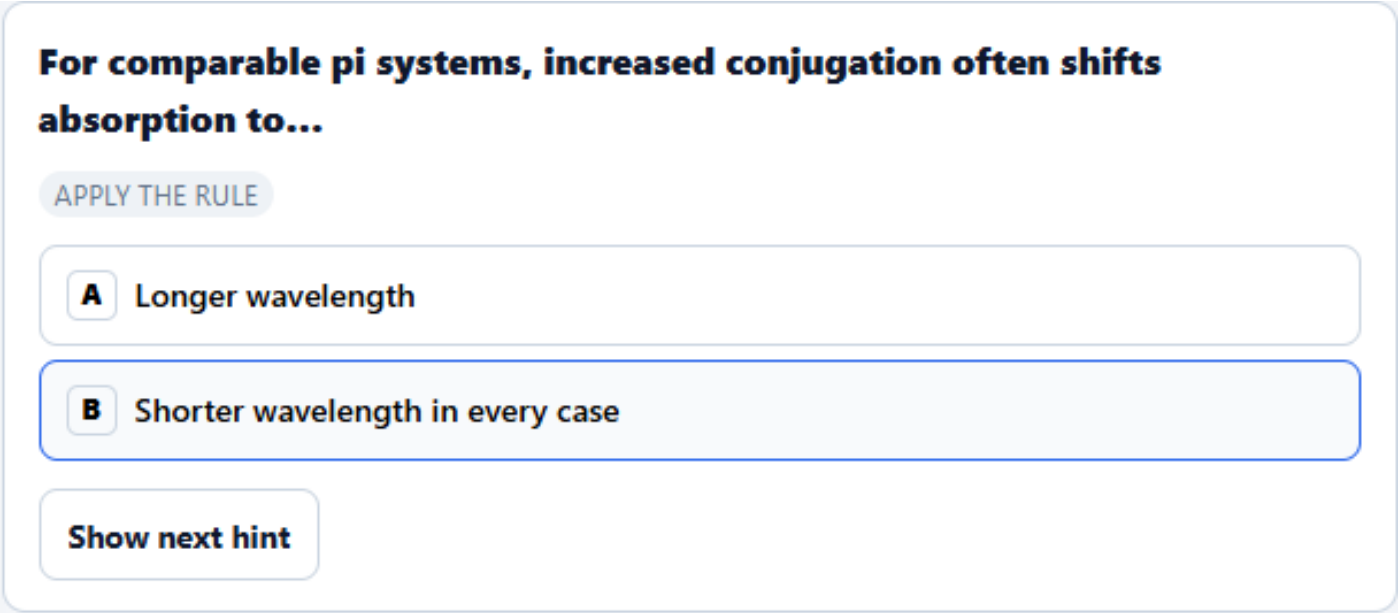
Option 0 (A): Longer wavelength

Option 1 (B): Shorter wavelength in every case

Screenshot: uv-energy-stage-2-molecule



Screenshot: uv-energy-stage-2-question



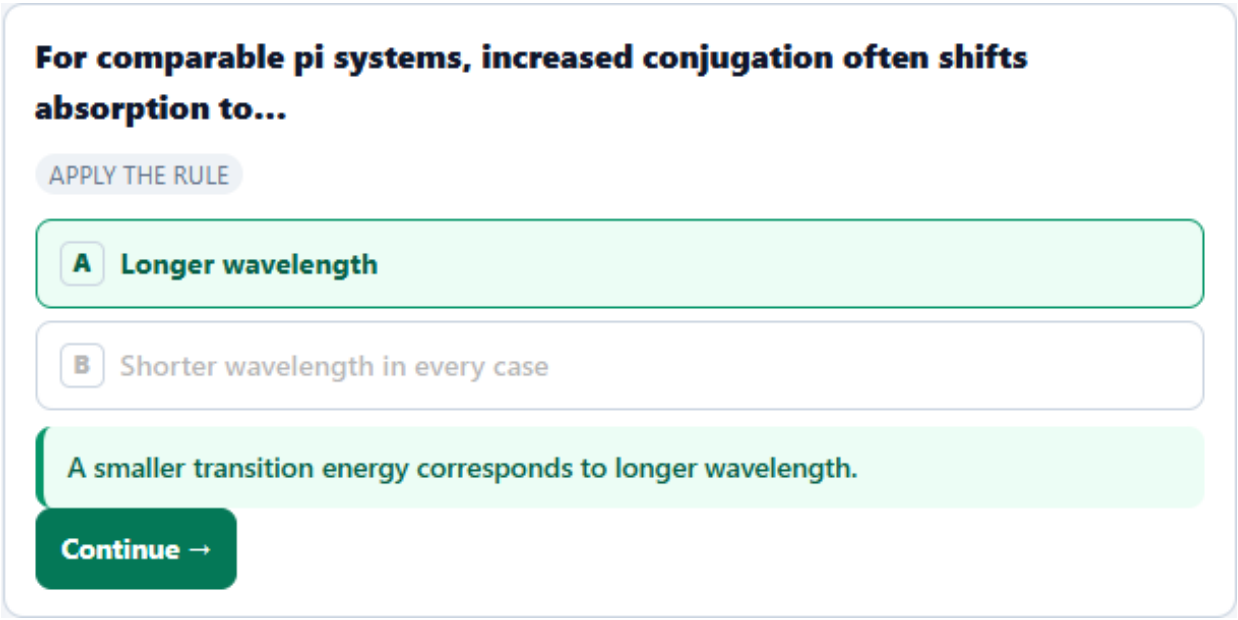
{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Use E=hc/λ.

Hint 2: Lower energy and longer wavelength are inversely related.

Explanation: A smaller transition energy corresponds to longer wavelength.

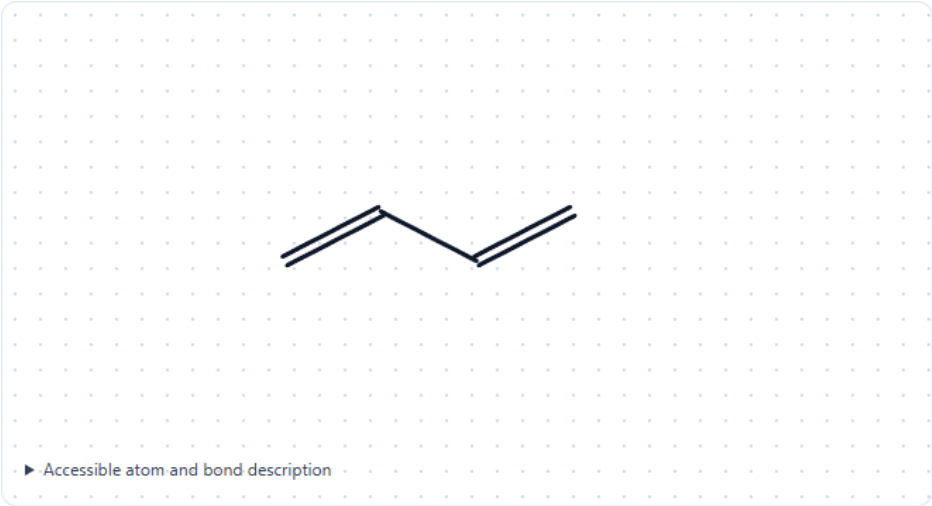
Screenshot: uv-energy-stage-2-feedback



uv-energy:stage-3 | If ε=10000 L mol^{-1} cm^{-1}, l=1 cm, and c=0.000020 mol L^{-1}, what is A?

Answer type: number | unit: dimensionless

Screenshot: uv-energy-stage-3-molecule



Screenshot: uv-energy-stage-3-question

If ε=10000 L mol⁻¹ cm⁻¹, l=1 cm, and c=0.000020 mol L⁻¹, what is A?

CALCULATE AND INTERPRET

Numeric answer (dimensionless)

Check answer

Show next hint

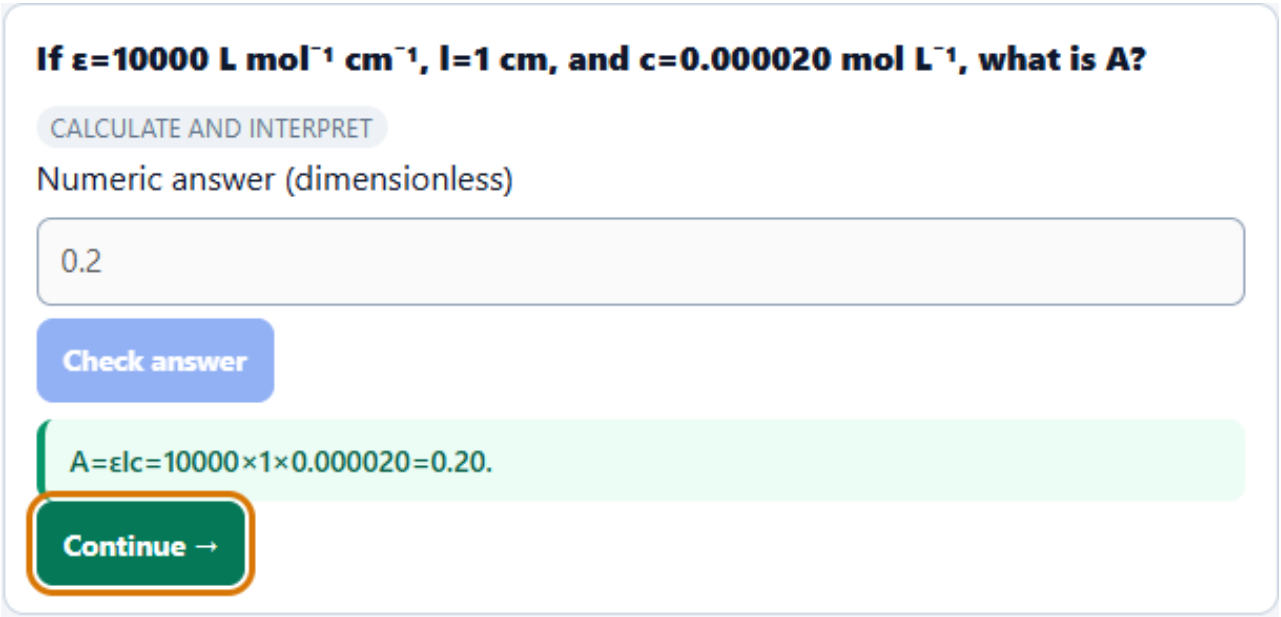
{"authorAnswer": 0.2, "tolerance": 0.001, "highlightsAfterCorrectAnswer": {}}

Hint 1: Multiply the three quantities.

Hint 2: The specified units cancel.

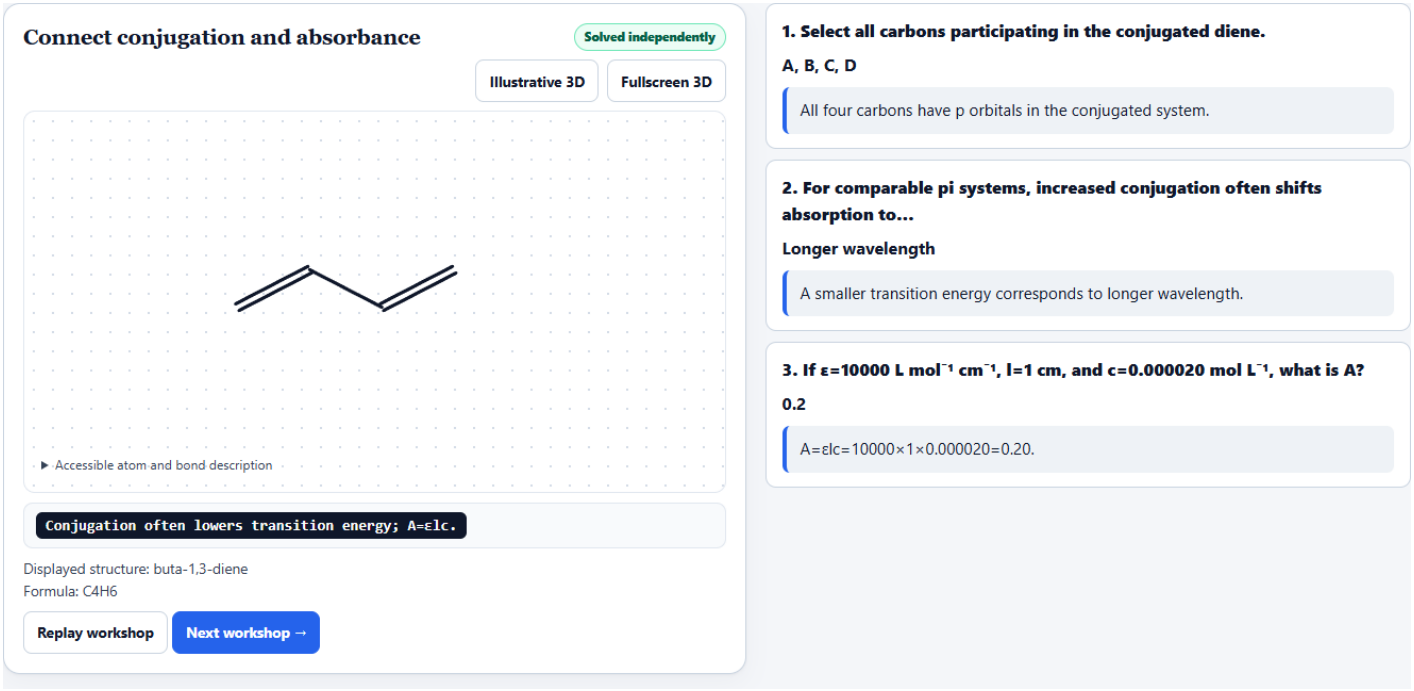
Explanation: $A=\varepsilon lc=10000\times1\times0.000020=0.20$.

Screenshot: uv-energy-stage-3-feedback



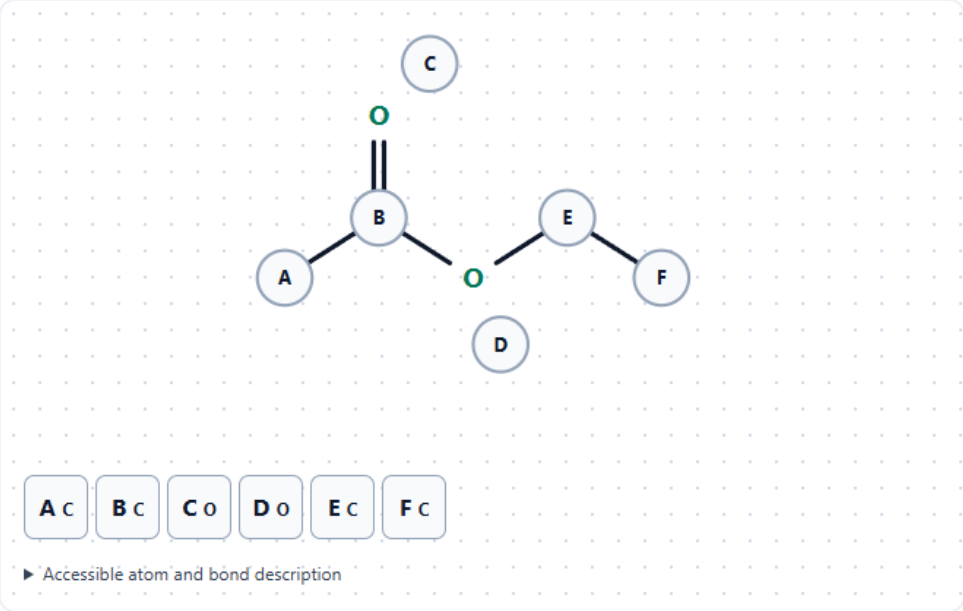
Completion summary: Conjugation often lowers transition energy; A=εlc.

Screenshot: uv-energy-complete

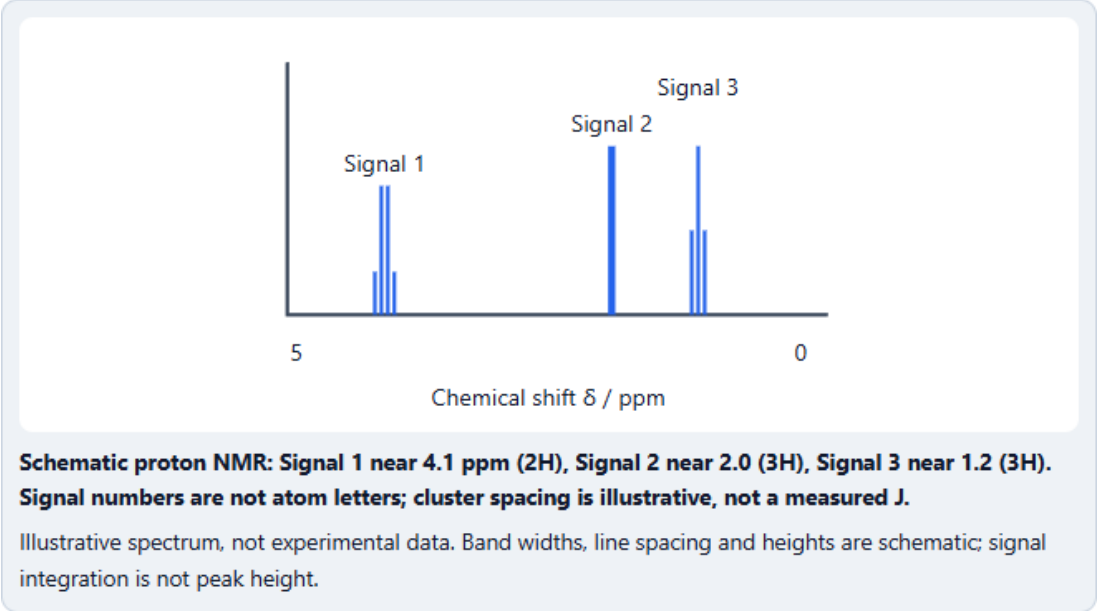


nmr-ethyl | Assign an ethyl-group pattern

Screenshot: nmr-ethyl-stage-1-molecule



Screenshot: nmr-ethyl-diagram

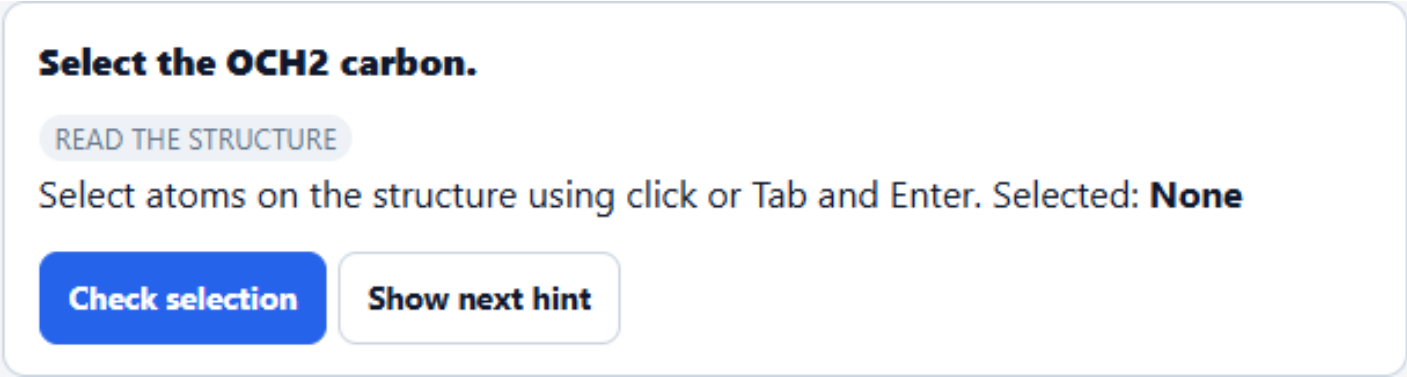


{"diagramSpecification": {"type": "spectrum", "min": 0, "max": 5, "unit": "Chemical shift δ / ppm", "caption": "Schematic proton NMR: Signal 1 near 4.1 ppm (2H), Signal 2 near 2.0 (3H), Signal 3 near 1.2 (3H). Signal numbers are not atom letters; cluster spacing is illustrative, not a measured J.", " peaks": [{"x": 4.1, "height": 0.65, "label": "Signal 1", "lines": 4, "spacing": 0.06}, {"x": 2, "height": 0.85, "label": "Signal 2", "lines": 1}, {"x": 1.2, "height": 0.85, "label": "Signal 3", "lines": 3, "spacing": 0.06}]}}

nmr-ethyl:stage-1 | Select the OCH2 carbon.

Answer type: atoms

Screenshot: nmr-ethyl-stage-1-question



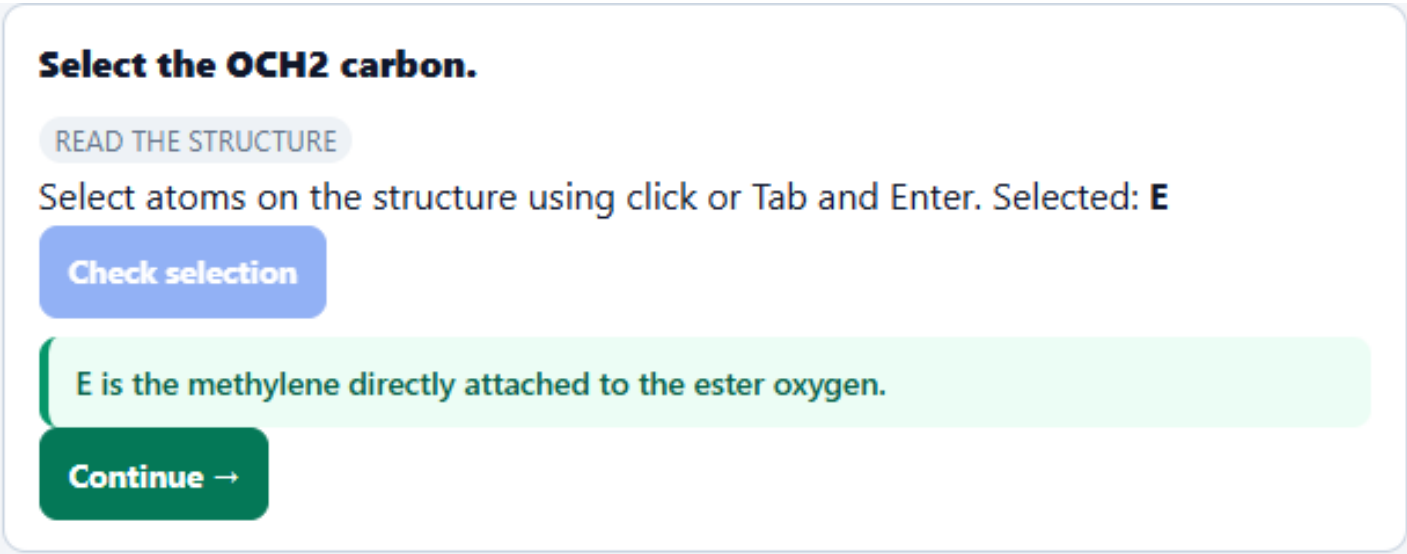
{"authorAnswer": [4], "tolerance": null, "highlightsAfterCorrectAnswer": {"prin": [4], "sub": [5]}}

Hint 1: Find O–CH2–CH3.

Hint 2: Choose the carbon nearest O on the ethyl side.

Explanation: E is the methylene directly attached to the ester oxygen.

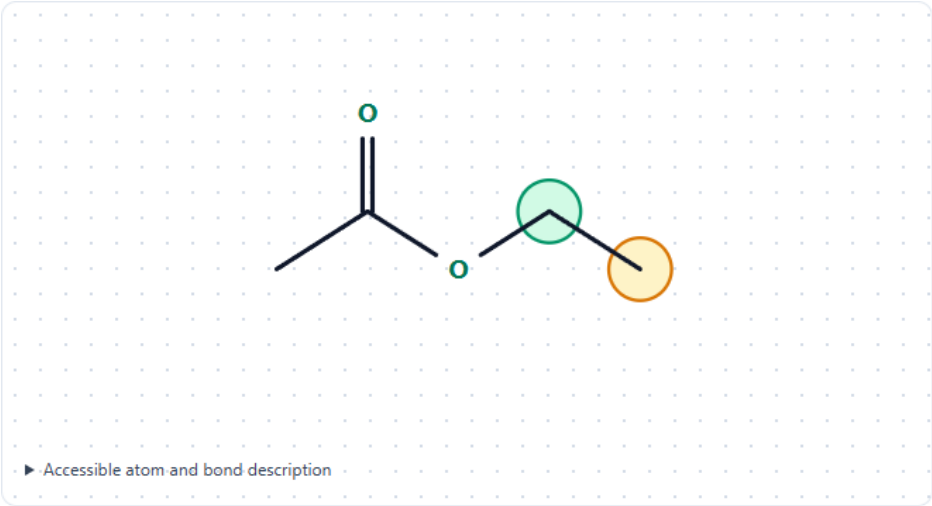
Screenshot: nmr-ethyl-stage-1-feedback



nmr-ethyl:stage-2 | How many lines does its signal have in the simple n+1 model?

Answer type: number | unit: lines

Screenshot: nmr-ethyl-stage-2-molecule



Screenshot: nmr-ethyl-stage-2-question

How many lines does its signal have in the simple n+1 model?

CALCULATE AND INTERPRET

Numeric answer (lines)

Check answer

Show next hint

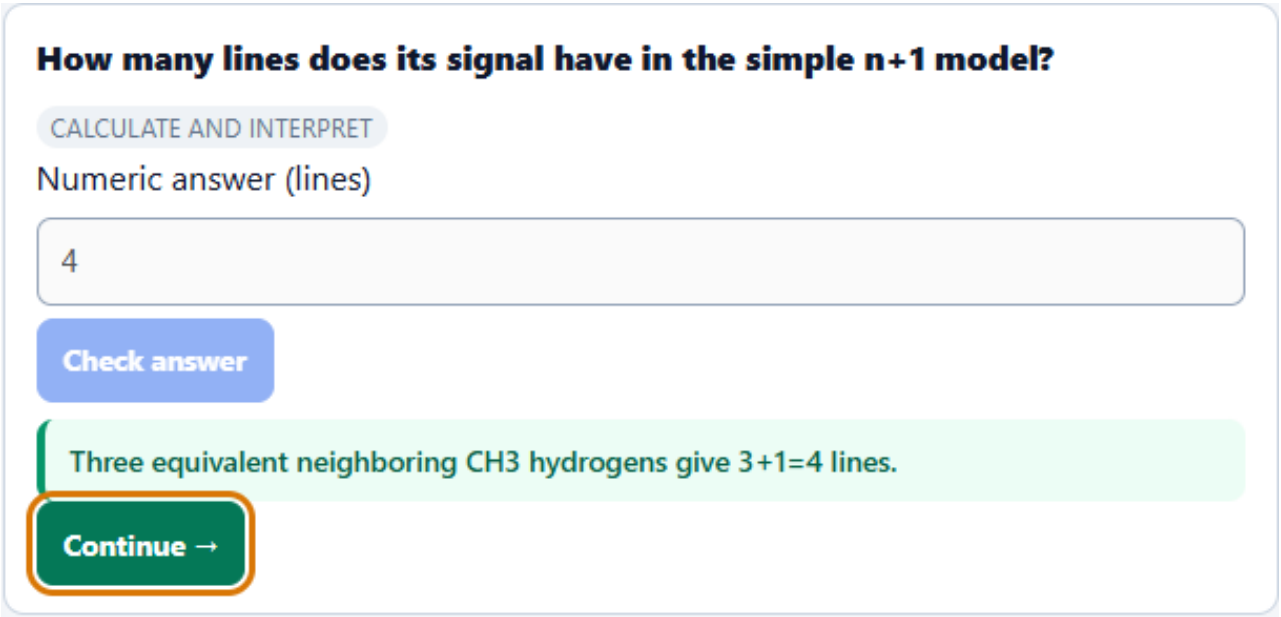
{"authorAnswer": 4, "tolerance": 0, "highlightsAfterCorrectAnswer": {}}

Hint 1: Count hydrogens on the adjacent methyl.

Hint 2: Do not count the two H on the observed carbon itself.

Explanation: Three equivalent neighboring CH3 hydrogens give 3+1=4 lines.

Screenshot: nmr-ethyl-stage-2-feedback



nmr-ethyl:stage-3 | The terminal ethyl CH3 gives an approximate…

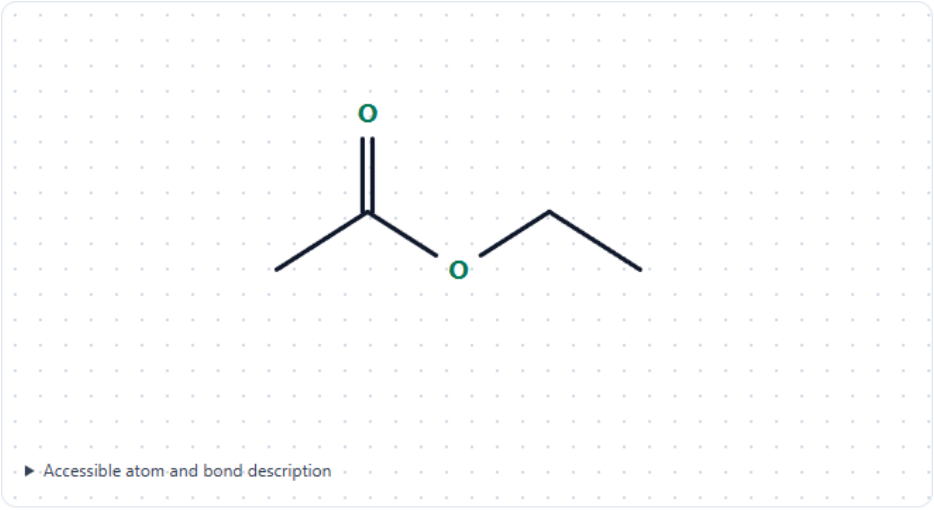
Answer type: choice

Option 0 (A): Triplet, integrating to 3H

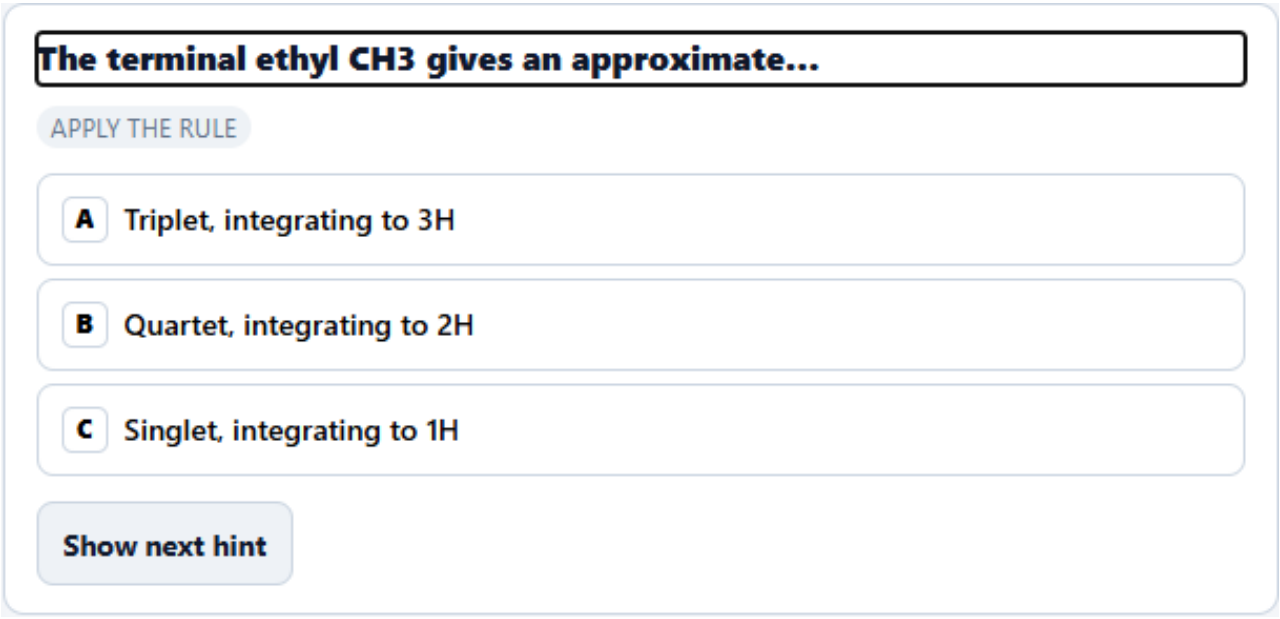
Option 1 (B): Quartet, integrating to 2H

Option 2 (C): Singlet, integrating to 1H

Screenshot: nmr-ethyl-stage-3-molecule



Screenshot: nmr-ethyl-stage-3-question



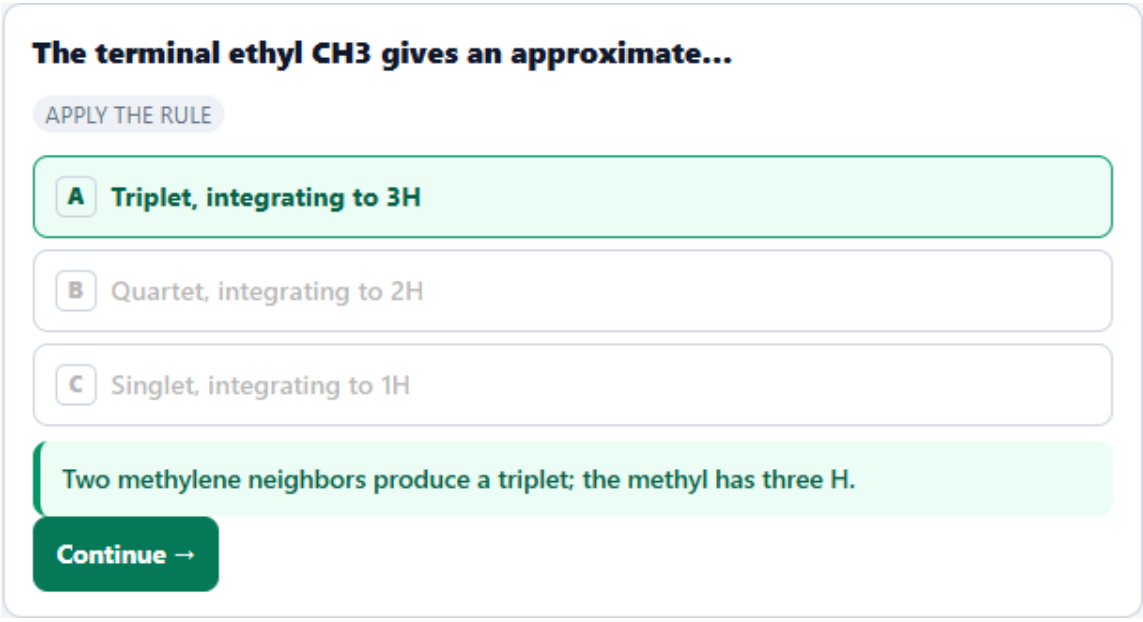
{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Splitting and integration are separate measurements.

Hint 2: Use n=2 for the neighbors.

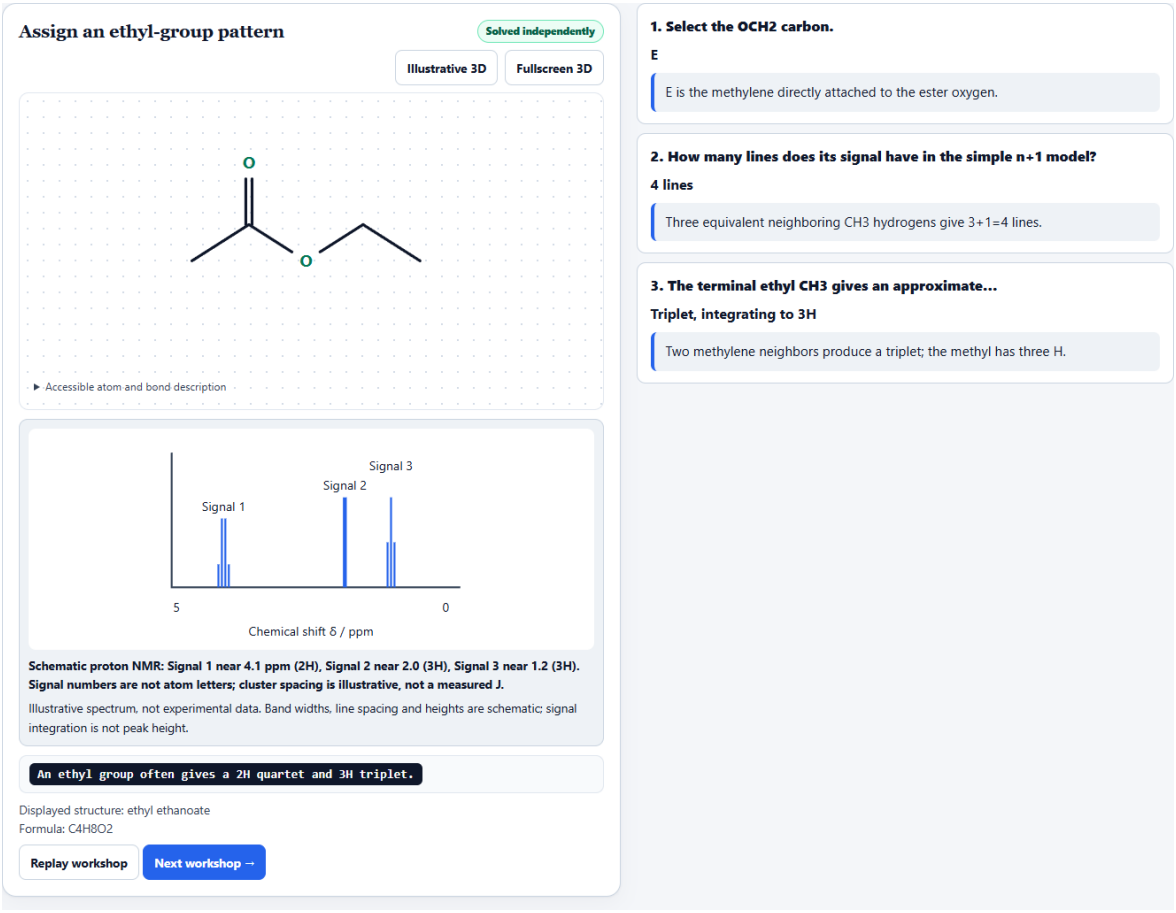
Explanation: Two methylene neighbors produce a triplet; the methyl has three H.

Screenshot: nmr-ethyl-stage-3-feedback



Completion summary: An ethyl group often gives a 2H quartet and 3H triplet.

Screenshot: nmr-ethyl-complete



combined-evidence | Test a structure against combined data

Supplied illustrative data: C3H6O; a ketone-like C=O band; one proton NMR singlet integrating to 6H near 2 ppm.

Screenshot: combined-evidence-stage-1-molecule

Candidate structure withheld. Use the supplied observations to infer it; the structure is shown after completion.

combined-evidence:stage-1 | For formula C3H6O, calculate DBE.

Answer type: number | unit: DBE

Screenshot: combined-evidence-stage-1-question

For formula C3H6O, calculate DBE.

CALCULATE AND INTERPRET

Numeric answer (DBE)

Check answer

Show next hint

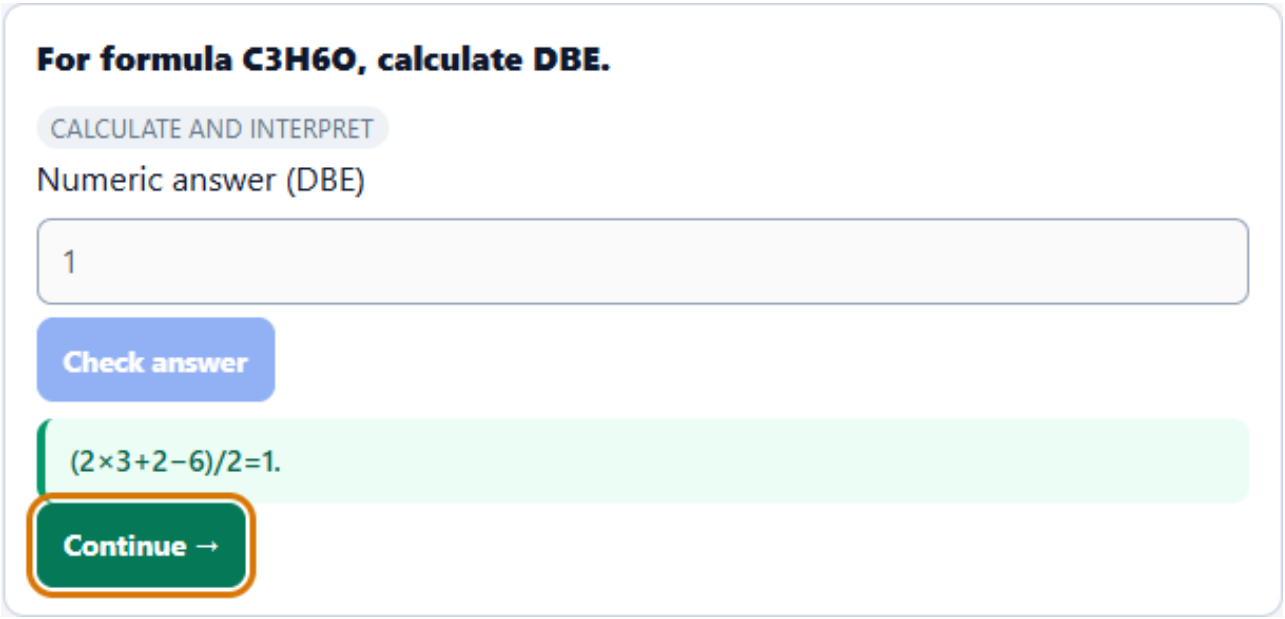
{"authorAnswer": 1, "tolerance": 0, "highlightsAfterCorrectAnswer": {}}

Hint 1: Ignore O in this standard formula.

Hint 2: Compare with the saturated acyclic C3H8 formula.

Explanation: (2×3+2−6)/2=1.

Screenshot: combined-evidence-stage-1-feedback



combined-evidence:stage-2 | A single 6H singlet near 2 ppm suggests...

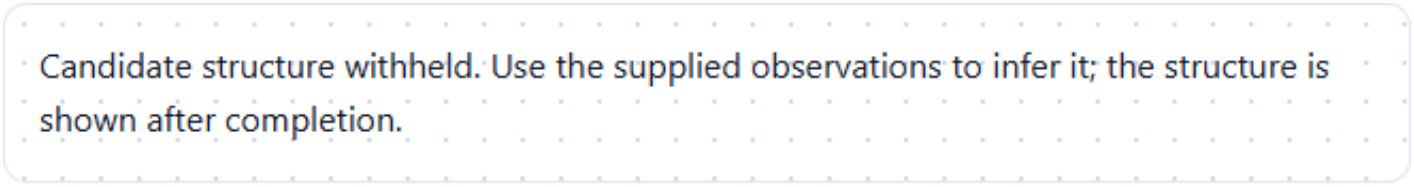
Answer type: choice

Option 0 (A): Two equivalent methyl groups with no adjacent H-bearing carbon

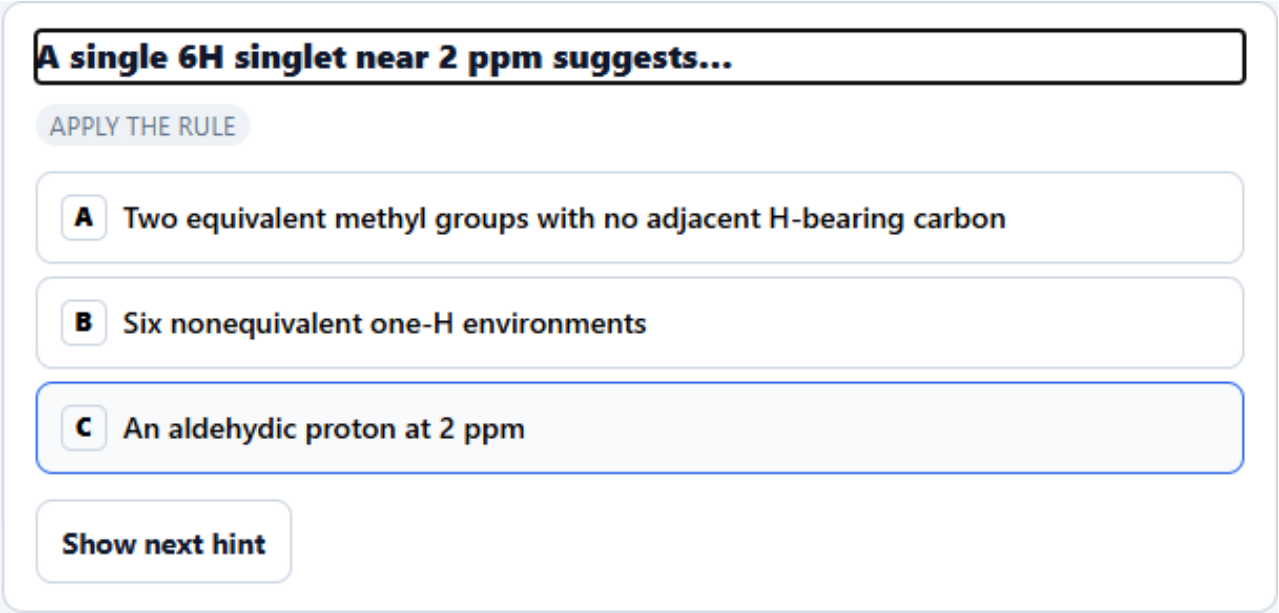
Option 1 (B): Six nonequivalent one-H environments

Option 2 (C): An aldehydic proton at 2 ppm

Screenshot: combined-evidence-stage-2-molecule



Screenshot: combined-evidence-stage-2-question



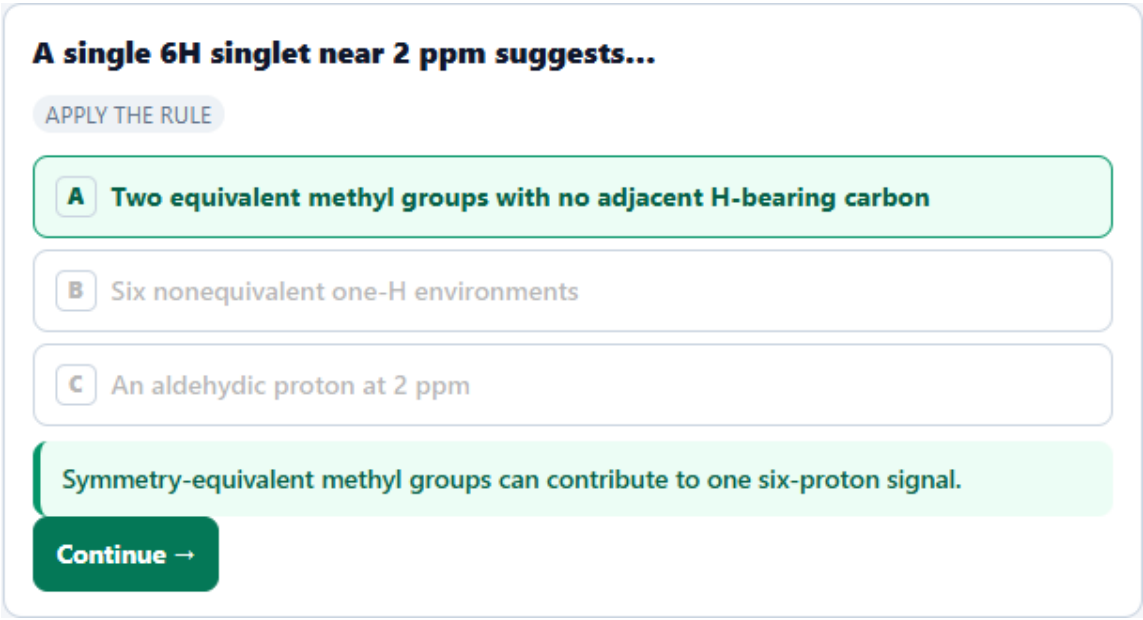
{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Integration counts all protons in the environment.

Hint 2: An aldehyde H would usually be much farther downfield.

Explanation: Symmetry-equivalent methyl groups can contribute to one six-proton signal.

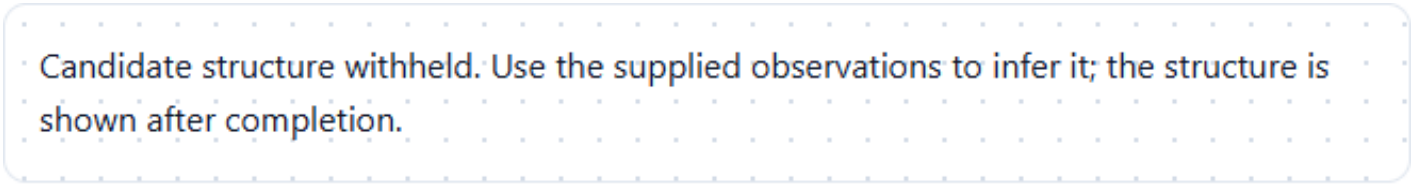
Screenshot: combined-evidence-stage-2-feedback



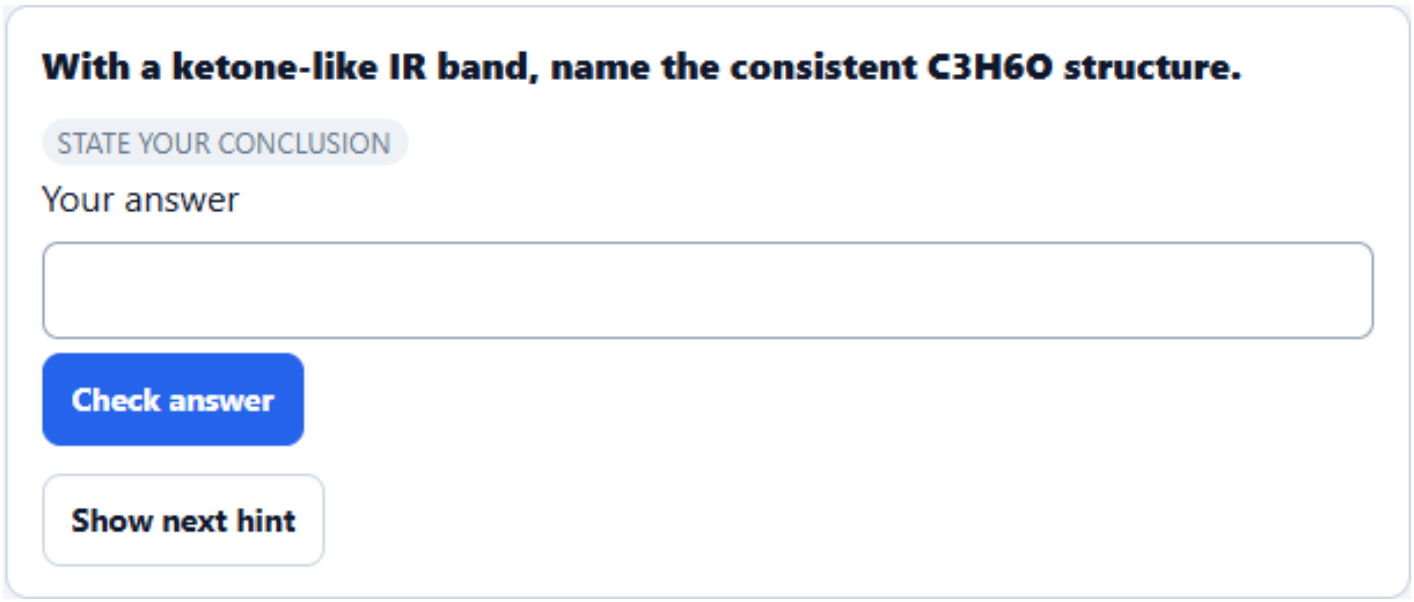
combined-evidence:stage-3 | With a ketone-like IR band, name the consistent C3H6O structure.

Answer type: text

Screenshot: combined-evidence-stage-3-molecule



Screenshot: combined-evidence-stage-3-question



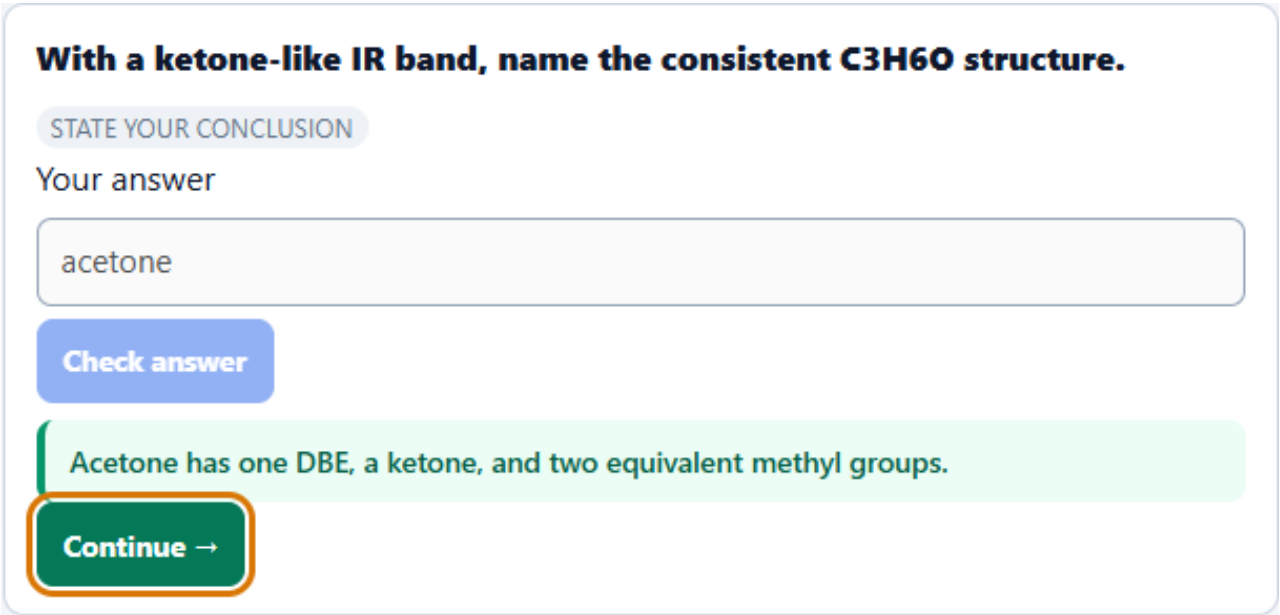
{"authorAnswer": ["acetone", "propanone", "propan-2-one", "2-propanone", "dimethyl ketone"], "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Place C=O between two methyl groups.

Hint 2: Check all three observations together.

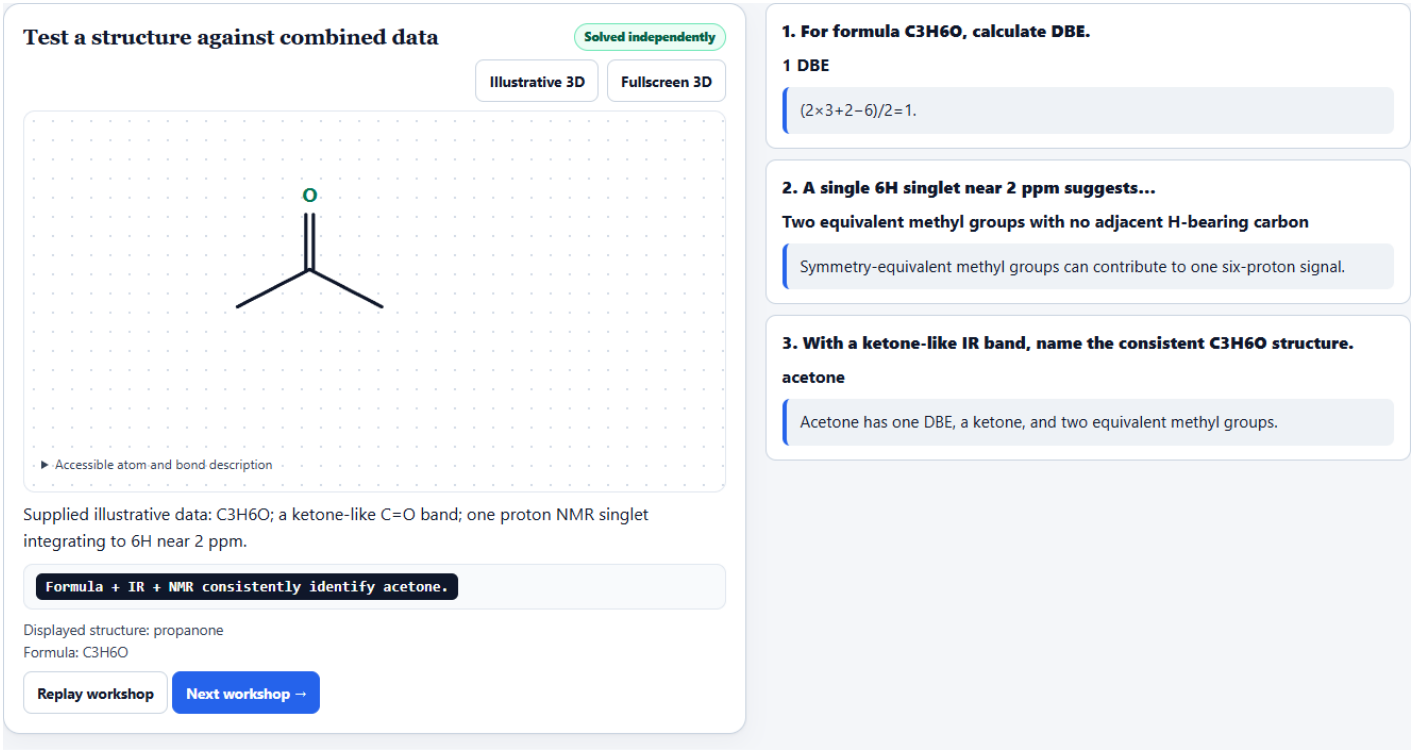
Explanation: Acetone has one DBE, a ketone, and two equivalent methyl groups.

Screenshot: combined-evidence-stage-3-feedback



Completion summary: Formula + IR + NMR consistently identify acetone.

Screenshot: combined-evidence-complete



Molecular repository: author claims to verify

acetone

Author claims: propanone | formula C3H6O | canonical composition C3H6O | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=50.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=108.00, y=85.00; charge=0; explicit H override=none

2 (C): C; x=166.00, y=115.00; charge=0; explicit H override=none

3 (D): O; x=108.00, y=25.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [1, 3, 2]]

isopropanol

Author claims: propan-2-ol | formula C3H8O | canonical composition C3H8O | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=50.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=108.00, y=85.00; charge=0; explicit H override=none

2 (C): C; x=166.00, y=115.00; charge=0; explicit H override=none

3 (D): O; x=108.00, y=25.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [1, 3, 1]]

aceticacid

Author claims: ethanoic acid | formula C2H4O2 | canonical composition C2H4O2 | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=50.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=108.00, y=85.00; charge=0; explicit H override=none

2 (C): O; x=108.00, y=25.00; charge=0; explicit H override=none

3 (D): O; x=166.00, y=115.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 2], [1, 3, 1]]

butadiene

Author claims: buta-1,3-diene | formula C4H6 | canonical composition C4H6 | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=50.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=108.00, y=85.00; charge=0; explicit H override=none

2 (C): C; x=166.00, y=115.00; charge=0; explicit H override=none

3 (D): C; x=224.00, y=85.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 2], [1, 2, 1], [2, 3, 2]]

ethene

Author claims: ethene | formula C2H4 | canonical composition C2H4 | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=131.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=189.00, y=85.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 2]]

ethylacetate

Author claims: ethyl ethanoate | formula C4H8O2 | canonical composition C4H8O2 | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=45.00, y=120.00; charge=0; explicit H override=none

1 (B): C; x=100.00, y=85.00; charge=0; explicit H override=none

2 (C): O; x=100.00, y=25.00; charge=0; explicit H override=none

3 (D): O; x=155.00, y=120.00; charge=0; explicit H override=none

4 (E): C; x=210.00, y=85.00; charge=0; explicit H override=none

5 (F): C; x=265.00, y=120.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 2], [1, 3, 1], [3, 4, 1], [4, 5, 1]]

propanal

Author claims: propanal | formula C3H6O | canonical composition C3H6O | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=95.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=155.00, y=85.00; charge=0; explicit H override=none

2 (C): C; x=215.00, y=115.00; charge=0; explicit H override=none

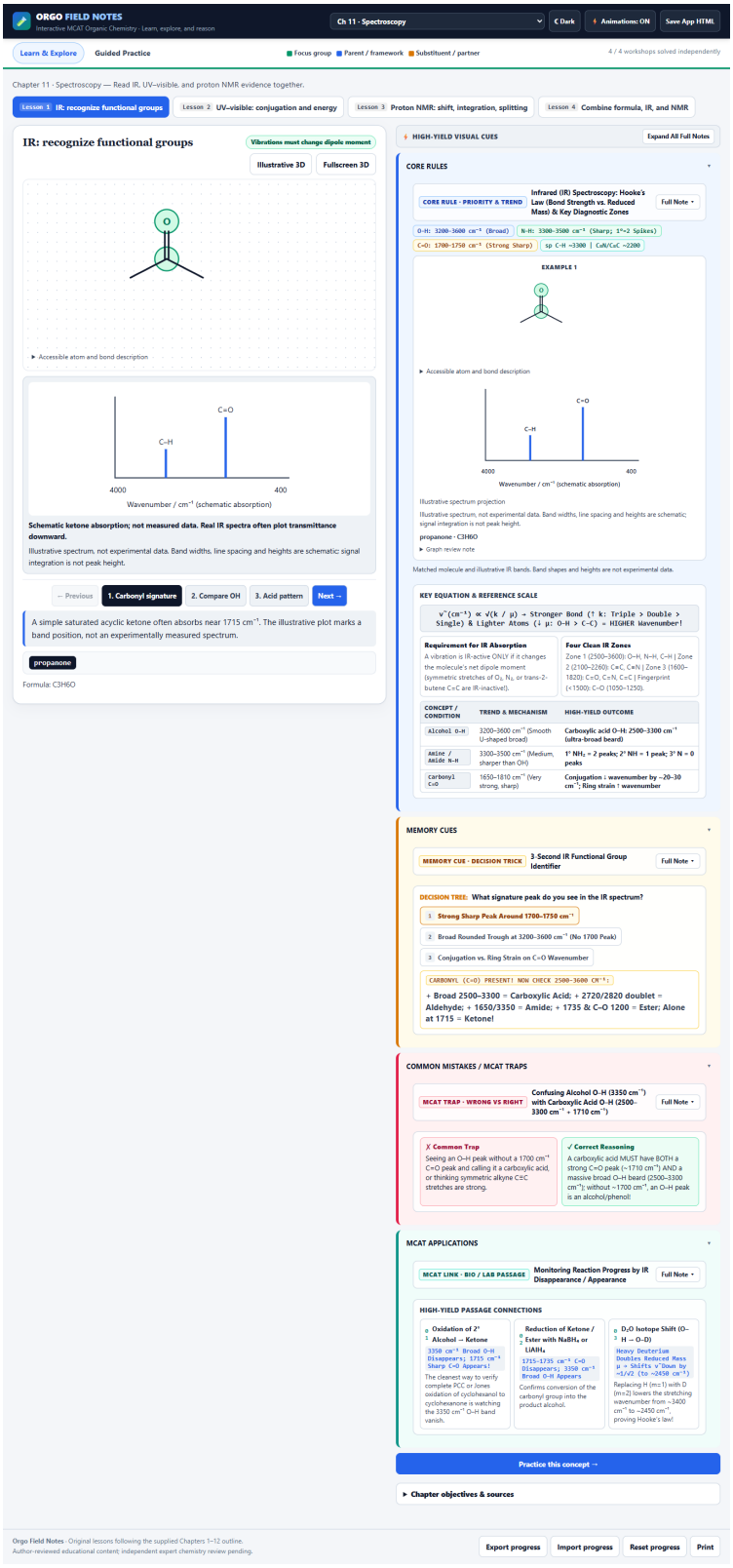
3 (D): O; x=267.00, y=85.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [2, 3, 2]]

Layout samples

These overview images are scaled for page fit. Detailed structure and stage images appear earlier. Samples cover the first concept on desktop and mobile in both themes, not every responsive state.

Screenshot: layout-desktop-light



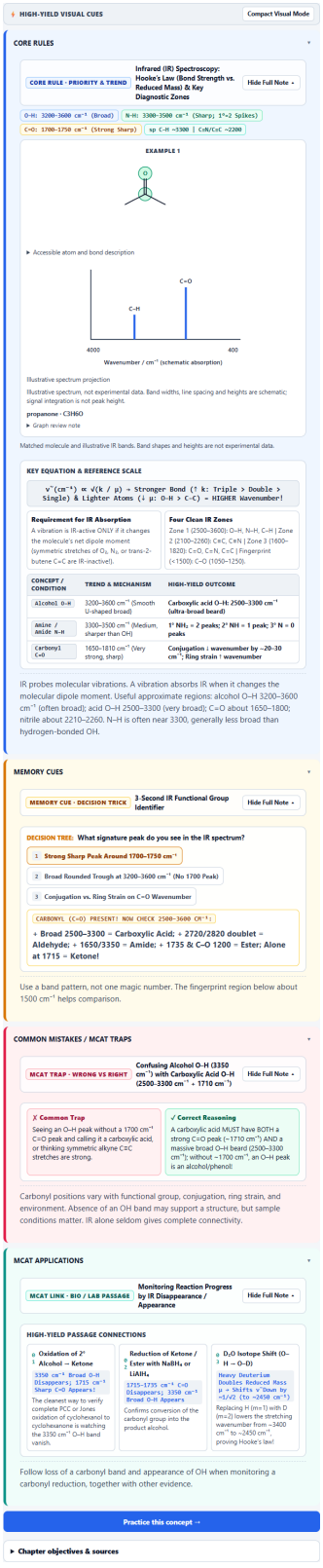
Screenshot: layout-mobile-light



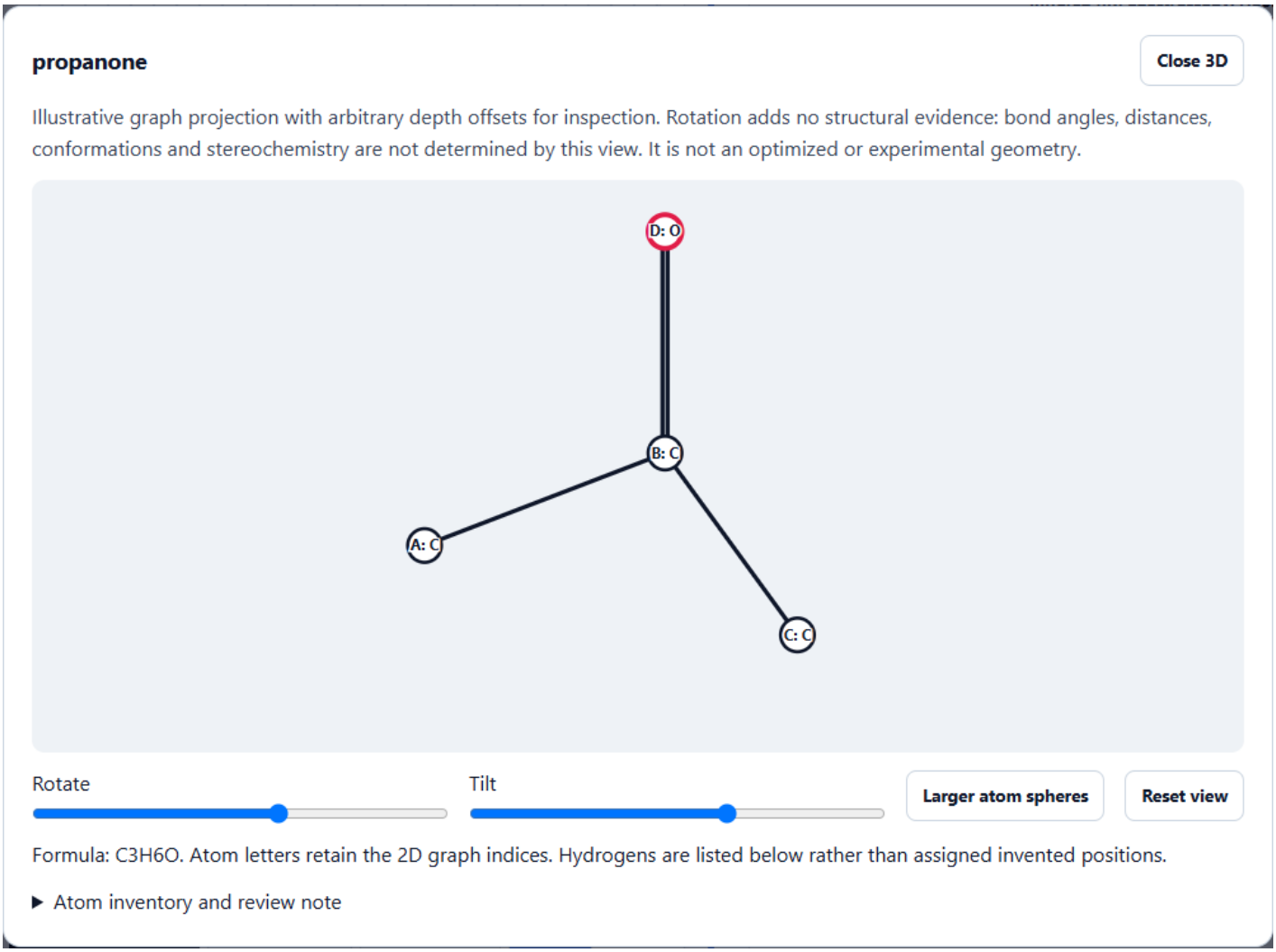
Screenshot: layout-mobile-dark



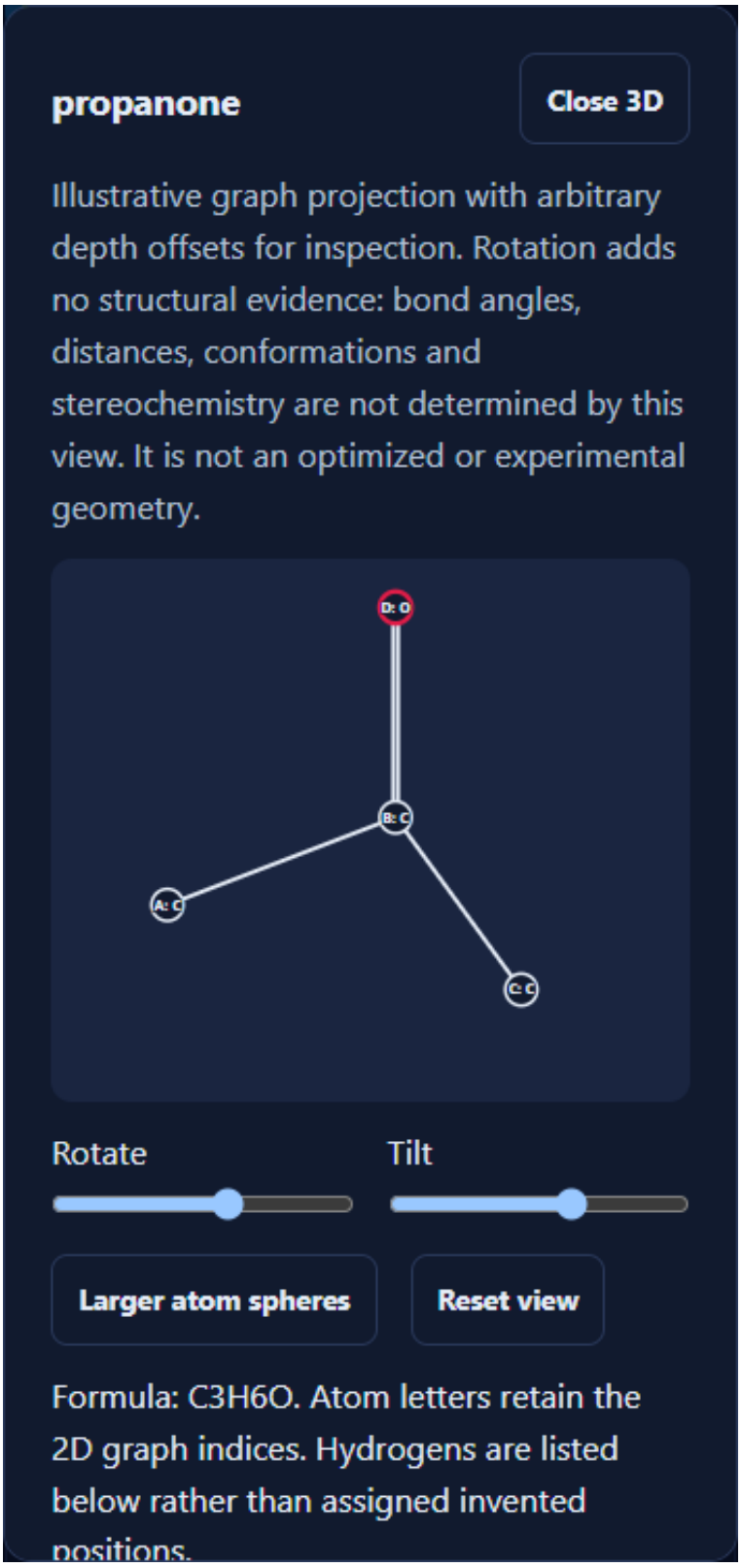
Screenshot: visual-notes-expanded



Screenshot: projection-desktop-light



Screenshot: projection-mobile-dark



Execution evidence

69 screenshots and 12 recorded interactions. Incorrect answers rejected and encoded keys accepted for captured guided stages. This checks implementation against its data; it does not validate chemistry independently. Static screenshots do not verify animation timing, arbitrary clicks, or screen-reader experience.